Proposed Subjects for the Ten Courses of the Faculty of Geoscience and Technology

Semester-by-semester subject plan with credits, syllabus, recommended textbooks and the most similar subjects at Tsinghua University, China University of Geosciences (Wuhan and Beijing), MIT and Colorado School of Mines

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Design principles

- 4-year Bachelor of Engineering: 8 semesters plus 3 summer field/practice terms, as at Tsinghua and CUG.
- 164 credits for every course - inside the 163-177 range of the Chinese benchmarks (Tsinghua 163-169, CUG Beijing 177) and above the US model (Mines 130 semester hours) because field schools and internships carry credit.
- Credit unit (assumed): 1 credit = 16 contact hours of lecture, 32 hours of laboratory, or 1 week of field practice/internship. Adjust if your national rules differ.
- Semesters 1-3 and part of Semester 4 are a common faculty foundation (75 credits) covering the fundamental subjects you listed: mathematics, physics, chemistry, material dynamics (engineering mechanics and mechanics of materials), fluid dynamics, electrical and electronic engineering, programming, plus physical geology and earth materials.
- Semesters 4-8 are course-specific (89 credits): Sem 4 (12), Summer 2 field school (4), Sem 5 (20), Sem 6 (20), Summer 3 internship (5), Sem 7 (18 incl. a team capstone design), Sem 8 (thesis 8 + frontiers seminar 2).
- Every course has a capstone design subject modelled on Mines' Mineral/Petroleum Exploration Design and Tsinghua's Integrated Course Design, and a professional English subject (CUGB model).

How to read the subject tables

- Syllabus: main topics to be covered; each subject still needs a full syllabus with weekly plan and assessment written by the responsible division.
- Textbooks: standard international textbooks (author, title, publisher). Use the latest edition. For design subjects, the national building/design code replaces foreign codes.
- Similar subjects: TH = Tsinghua 2023 plan course code; CUGB = CUG (Beijing) Resources Prospecting Engineering plan (semester shown); CUG-W = CUG (Wuhan) programme; MIT = MIT subject number; Mines = Colorado School of Mines course code. Where a benchmark offers the topic only as a programme or research centre, the programme is named. Where none of the five offers it, the table says so.

A. Overview: credits by semester for each course

Course (responsible institute)	Sem 1	Sem 2	Summer 1	Sem 3	Sem 4	Summer 2	Sem 5	Sem 6	Summer 3	Sem 7	Sem 8	Total
Common faculty foundation (all courses)	22	22	2	21	8							75
Mineral Exploration - course part					12	4	20	20	5	18	10	89
Civil Engineering - course part					12	4	20	20	5	18	10	89
Geophysical Exploration - course part					12	4	20	20	5	18	10	89
Geochemical Exploration - course part					12	4	20	20	5	18	10	89
Drill Engineering - course part					12	4	20	20	5	18	10	89
Mineral Engineering - course part					12	4	20	20	5	18	10	89
Geological Information - course part					12	4	20	20	5	18	10	89
Space - course part					12	4	20	20	5	18	10	89
Tourist District Development - course part					12	4	20	20	5	18	10	89
Petroleum Exploration and Engineering - course part					12	4	20	20	5	18	10	89
Semester total (every course)	**22**	**22**	**2**	**21**	**20**	**4**	**20**	**20**	**5**	**18**	**10**	**164**

Credit blocks per course

Block	Credits	Share	Contents
General education	9	5%	Languages, humanities, PE (Sem 1-4) - per national rules
Mathematics and natural science	31	19%	Calculus I-II, linear algebra, differential equations, probability and statistics, physics I-II + lab, chemistry
Engineering and computing fundamentals	24	15%	Programming, engineering graphics, engineering mechanics, mechanics of materials, fluid mechanics, electrical and electronic engineering, numerical methods
Geoscience fundamentals and orientation	11	7%	Introduction seminar, physical geology, earth materials, recognition field practice
Course-specific subjects incl. capstone (Sem 4-7)	70	43%	See sections 1-10
Field school and internship	9	5%	Summer 2 field school (4), Summer 3 internship (5)
Thesis and frontiers seminar	10	6%	Sem 8
Total	164	100%	

Course list and responsible institutes

No.	Course	Responsible institute / division	Graduate profile
1	Mineral Exploration	Institute of Resource Exploration (geology, ore deposit divisions)	Exploration geologists able to generate targets, run mapping, geochemical, geophysical and drilling programmes, estimate resources and report to international standards.
2	Civil Engineering	Institute of Civil Engineering (civil building and prospecting, civil building and engineering divisions)	Civil engineers who can investigate ground conditions, design concrete and steel structures, foundations and underground works, and manage construction.
3	Geophysical Exploration	Institute of Geophysical Prospecting (electromagnetic, magnetic, prospecting-equipment divisions)	Geophysicists who can design, acquire, process and interpret gravity, magnetic, electrical, EM and seismic surveys, and develop field instruments.
4	Geochemical Exploration	Institute of Resource Exploration (geochemical prospecting division)	Exploration and environmental geochemists who can plan surveys, run and control laboratory analyses, process multivariate data and interpret anomalies.

No.	Course	Responsible institute / division	Graduate profile
5	Drill Engineering	Independent Drill Engineering Division	Drilling engineers for mineral-exploration, water-well, geotechnical, geothermal and petroleum drilling: method selection, rig and bit design, fluids, well design and safety.
6	Mineral Engineering	Institute of Mineral Engineering (mineralogy and petrology, mineral material development, rock application divisions)	Mineral engineers who characterise minerals and rocks, process them, and develop industrial mineral, stone, aggregate and functional-material products.
7	Geological Information	Laboratory of Space (geospatial information division) with the Institute of Resource Exploration	Geo-information engineers who build geological databases, GIS and 3D models, process remote-sensing data and apply machine learning to exploration and hazards.
8	Space	Laboratory of Space (space science, earth atmosphere, geospatial information divisions)	Space and atmospheric scientists and engineers who can work on Earth-observation satellites, orbital and spacecraft systems, weather and climate, and space weather.
9	Tourist District Development	Institute of Civil Engineering (tourist district development division)	Planners and engineers who assess landscapes and geoheritage, prepare tourist-district master plans, design facilities and infrastructure and manage destinations sustainably.
10	Petroleum Exploration and Engineering	Institute of Petrol (prospecting, mining engineering, mining machine, processing divisions)	Petroleum geoscientists and engineers covering exploration, drilling, reservoir and production engineering, equipment and refining.

B. Common faculty foundation (Semesters 1-4, all ten courses, 75 credits)

All students take these subjects. In Semester 4 they add 12 credits from their own course (sections 1-10).

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Sem 1 22 cr total	**General education (languages, humanities, physical education)**	3	Per national/university regulations: foreign language, history and social sciences, physical education, writing and communication.	Set by the university	TH university general education (47 cr); CUGB general education (39 cr required + 16 elective)
	Calculus I	5	Limits and continuity; derivatives and applications (rates, optimisation); integrals, techniques of integration; applications to areas, volumes, work.	Stewart, Calculus: Early Transcendentals (Cengage)	TH 10421055 Calculus A(1); CUGB Advanced Mathematics A1 (S1); Mines MATH111; MIT 18.01
	Linear Algebra	3	Linear systems and Gaussian elimination; matrices and determinants; vector spaces; eigenvalues/eigenvectors; orthogonality and least squares.	Strang, Introduction to Linear Algebra (Wellesley-Cambridge Press)	TH 10421324 Linear Algebra; CUGB Linear Algebra (S3); MIT 18.06
	General Physics I (Mechanics, Heat, Waves)	4	Kinematics, Newton's laws, energy and momentum, rotation, gravitation, oscillations and waves, fluids, temperature and the laws of thermodynamics.	Halliday, Resnick and Walker, Fundamentals of Physics (Wiley)	TH 10431174 University Physics CE; CUGB College Physics (1); Mines PHGN100; MIT 8.01
	General Chemistry with Laboratory	3	Atomic structure, bonding, stoichiometry, thermochemistry, chemical equilibrium, acids-bases, redox, solution chemistry; lab techniques and titration.	Brown, LeMay, Bursten et al., Chemistry: The Central Science (Pearson)	TH 10440012 University Chemistry B; CUGB College Chemistry + Chemistry Experiments; Mines CHGN121/122
	Programming for Engineers (Python)	3	Variables, control flow, functions, files; NumPy and pandas; plotting with Matplotlib; scripting for field and laboratory data.	Downey, Think Python (O'Reilly); McKinney, Python for Data Analysis (O'Reilly)	TH 30250023 Computer Language and Programming; CUGB Computer Language Programming; Mines CSCI128; MIT 6.100A, 12.010
	Introduction to Geoscience and Technology (seminar)	1	Overview of all ten courses, careers, professional ethics, field and laboratory safety, visits to the faculty's institutes.	Faculty lecture notes	TH 20030181 Intro to Civil, Hydraulic and Ocean Eng.; CUGB Intro to the profession; Mines GEOL102; MIT 12.00 Frontiers and Careers
Sem 2 22 cr total	**General education (languages, humanities, physical education)**	2	Per national/university regulations: foreign language, history and social sciences, physical education, writing and communication.	Set by the university	TH university general education (47 cr); CUGB general education (39 cr required + 16 elective)
	Calculus II (Multivariable and Vector Calculus)	5	Sequences and series; partial derivatives; multiple integrals; vector fields, gradient, divergence, curl; line and surface integrals; Green, Stokes and Gauss theorems.	Stewart, Calculus: Early Transcendentals (Cengage)	TH 10421065 Calculus A(2); CUGB Advanced Mathematics A2 (S2); Mines MATH112/MATH213; MIT 18.02
	General Physics II (Electromagnetism, Optics, Modern Physics)	4	Electric fields and potential, circuits, magnetism, induction, Maxwell's equations, EM waves, optics, introduction to atomic and nuclear physics (radioactivity).	Halliday, Resnick and Walker, Fundamentals of Physics (Wiley)	CUGB College Physics (2) (S3); Mines PHGN200; MIT 8.02
	Physics Laboratory	1	Measurement and error analysis; experiments in mechanics, heat, electricity, magnetism and optics.	University laboratory manual	TH 10430782 Physics Experiments A(1); CUGB Physics Experiments (S3)
	Physical Geology (with laboratory)	4	Earth structure, plate tectonics, minerals and rocks, magmatism, weathering and surface processes, geologic time, geologic maps, resources and hazards.	Marshak, Earth: Portrait of a Planet (W.W. Norton); Grotzinger and Jordan, Understanding Earth (W.H. Freeman)	CUGB Geosciences (S2); Mines GEGN101 Earth and Environmental Systems; MIT 12.001 Introduction to Geology
	Engineering Mechanics (Statics and Dynamics)	4	Force systems, equilibrium of particles and rigid bodies, trusses and frames, friction, centroids and moments of inertia; kinematics and kinetics, work-energy, impulse-momentum.	Hibbeler, Engineering Mechanics: Statics and Dynamics (Pearson)	TH 20310334 Theoretical Mechanics; Mines CEEN241 Statics
	Engineering Graphics and CAD	2	Orthographic and isometric projection, sections, dimensioning, standards; 2D/3D CAD; drafting of geological sections and engineering drawings.	Giesecke et al., Technical Drawing with Engineering Graphics (Pearson)	TH 20120152 Fundamentals of Engineering Graphics + 20900012 Engineering Computer Drafting; Mines EDNS151 Cornerstone Design I
Summer 1 2 cr total	**Geological Recognition Field Practice (2 weeks)**	2	Field observation of rocks, structures and landforms; compass-clinometer and GNSS use; field notebook; sketching outcrops.	Coe (ed.), Geological Field Techniques (Wiley-Blackwell)	CUGB Beidaihe Geological Field Trip (2 cr); TH 40030711 Cognition Practice; Mines GEGN217 Geologic Field Methods
Sem 3 21 cr total	**General education (languages, humanities, physical education)**	2	Per national/university regulations: foreign language, history and social sciences, physical education, writing and communication.	Set by the university	TH university general education (47 cr); CUGB general education (39 cr required + 16 elective)

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
	Differential Equations and Mathematical Physics	3	First/second-order ODEs, systems, Laplace transform, Fourier series; heat, wave and Laplace equations with applications to diffusion, groundwater and potential fields.	Kreyszig, Advanced Engineering Mathematics (Wiley)	Mines MATH225 Differential Equations; MIT 18.03; TH 10420262 Intro to Mathematical-Physical Equations
	Probability and Statistics	3	Probability, random variables, distributions, sampling, estimation, hypothesis tests, regression and correlation, introduction to spatial sampling.	Montgomery and Runger, Applied Statistics and Probability for Engineers (Wiley)	TH 10420803 Probability and Statistics; CUGB Probability and Mathematical Statistics A (S3); MIT 12.012 (part)
	Mechanics of Materials	3	Stress and strain, axial load, torsion, bending, shear, combined stresses, Mohr's circle, deflection of beams, buckling of columns.	Hibbeler, Mechanics of Materials (Pearson)	TH 20310343 Mechanics of Materials; Mines CEEN311 Mechanics of Materials
	Fluid Mechanics	3	Fluid properties, hydrostatics, continuity, Bernoulli and momentum equations, dimensional analysis, pipe flow, open-channel flow, flow through porous media (Darcy's law).	White, Fluid Mechanics (McGraw-Hill)	TH 30040493 Fluid Mechanics; Mines GEGN351 Geological Fluid Mechanics
	Electrical Engineering Fundamentals	3	DC and AC circuit analysis, phasors, three-phase power, transformers, electric motors and generators, electrical safety on sites.	Hambley, Electrical Engineering: Principles and Applications (Pearson)	No direct equivalent in the five benchmark plans reviewed (covered through physics at Mines/MIT)
	Earth Materials (Mineralogy and Petrology I)	4	Crystallography and crystal chemistry, rock-forming and ore minerals, hand-specimen identification, classification of igneous, sedimentary and metamorphic rocks, intro to the polarising microscope.	Klein and Philpotts, Earth Materials: Introduction to Mineralogy and Petrology (Cambridge)	CUGB Crystallography and Mineralogy (S3) + Crystal Optics; Mines GEGN212 The Rock Cycle; MIT 12.108 Earth Materials
Sem 4 **8 cr total**	**General education (languages, humanities, physical education)**	2	Per national/university regulations: foreign language, history and social sciences, physical education, writing and communication.	Set by the university	TH university general education (47 cr); CUGB general education (39 cr required + 16 elective)
	Electronic Engineering and Instrumentation	3	Diodes, transistors, operational amplifiers, filters, ADC/DAC, sensors, microcontrollers and data acquisition for field instruments.	Hambley, Electrical Engineering: Principles and Applications (Pearson); Horowitz and Hill, The Art of Electronics (Cambridge)	No direct equivalent in the five benchmark plans reviewed; related to CUGB Dept. of Measurement and Control Technology
	Numerical Methods and Geodata Analysis	3	Root finding, interpolation, numerical integration, finite differences for ODE/PDE, optimisation, time series and introductory machine learning in Python.	Chapra and Canale, Numerical Methods for Engineers (McGraw-Hill); Trauth, Python/MATLAB Recipes for Earth Sciences (Springer)	MIT 12.010 / 12.012; Mines GEGN432 Geological Data Management; TH 40250443 Numerical Analysis and Algorithms, 30030622 Math Modelling and Data Analysis

1. Mineral Exploration course

Responsible unit: Institute of Resource Exploration (geology, ore deposit divisions). Graduate profile: Exploration geologists able to generate targets, run mapping, geochemical, geophysical and drilling programmes, estimate resources and report to international standards.

Course-specific credits: Sem 4 12, Summer 2 4, Sem 5 20, Sem 6 20, Summer 3 5, Sem 7 18, Sem 8 10 = 89. Semesters 1-3 and the common part of Sem 4 (8 cr): see section B. Total degree: 164 credits.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Sem 4 12 cr total	**Petrology and Optical Petrography**	4	Optical mineralogy; igneous, metamorphic and sedimentary petrology; thin-section description; phase diagrams; rocks as records of tectonic setting.	Philpotts and Ague, Principles of Igneous and Metamorphic Petrology (Cambridge); Nesse, Introduction to Optical Mineralogy (Oxford)	CUGB Petrology (S4, 5.5 cr); Mines GEGN307 Petrology; MIT 12.108
	Structural Geology	4	Stress and strain, faults, folds, foliations, shear zones, stereographic projection, structural analysis of maps and cross-sections; structural control of ore and traps.	Fossen, Structural Geology (Cambridge); Lisle, Geological Structures and Maps (Elsevier)	CUGB Structural Geology (S4); Mines GEOL309; MIT 12.113
	Paleontology, Stratigraphy and Historical Geology	4	Fossil groups and biostratigraphy, lithostratigraphy, chronostratigraphy, sequence stratigraphy basics, Earth history and regional geology.	Stanley and Luczaj, Earth System History (W.H. Freeman); Boggs, Principles of Sedimentology and Stratigraphy (Pearson)	CUGB Paleontology + Historical Geology (S4); Mines GEOL314 Stratigraphy, GEOL444
Summer 2 4 cr total	**Geological Mapping Field School (4 weeks)**	4	Systematic mapping of a sedimentary-igneous-metamorphic terrain; measured sections; structural data; geological map, cross-sections and field report.	Lisle, Brabham and Barnes, Basic Geological Mapping (Wiley-Blackwell)	CUGB Zhoukoudian Mapping Field School (6 wks); Mines GEGN316 Field Geology (5 wks); MIT 12.115
Sem 5 20 cr total	**Ore Deposit Geology**	4	Ore-forming processes: magmatic, hydrothermal, sedimentary, supergene; deposit models (porphyry, VMS, epithermal, orogenic gold, skarn, SEDEX, IOCG, REE); ore mineral assemblages.	Robb, Introduction to Ore-Forming Processes (Wiley-Blackwell); Pohl, Economic Geology: Principles and Practice (Wiley-Blackwell)	CUGB Ore Deposit Geology (S5, 4 cr); Mines GEGN401 Mineral Deposits; CUG-W Ore Deposit Geology Field Training
	Geochemistry	3	Element distribution and cosmochemistry, crystal chemistry, thermodynamics of geologic systems, aqueous geochemistry, trace elements, isotope basics, organic geochemistry introduction.	White, Geochemistry (Wiley-Blackwell); Misra, Introduction to Geochemistry: Principles and Applications (Wiley-Blackwell)	CUGB Geochemistry incl. organic (S5); Mines CHGN406 Intro to Geochemistry; MIT 12.104
	Ore Microscopy and Ore Petrography	3	Reflected-light microscopy, identification of ore minerals, ore textures and paragenesis, application to beneficiation.	Craig and Vaughan, Ore Microscopy and Ore Petrography (Wiley); Ramdohr, The Ore Minerals and Their Intergrowths (Pergamon)	CUGB Ore Petrology and Microscopy (S5); Mines GEGN404 Ore Microscopy
	Applied Geophysics for Exploration	3	Gravity, magnetic, electrical, electromagnetic, radiometric and seismic methods; survey design and interpretation for mineral exploration.	Dentith and Mudge, Geophysics for the Mineral Exploration Geoscientist (Cambridge); Kearey, Brooks and Hill, An Introduction to Geophysical Exploration (Wiley-Blackwell)	CUGB Exploration Geophysics Methods and Applications (S5); MIT 12.002, 12.211; Mines GEGN439 (integration)
	Remote Sensing Geology	2	EM spectrum, optical and radar sensors, image enhancement, lithological and alteration mapping, structural lineament analysis.	Gupta, Remote Sensing Geology (Springer); Lillesand, Kiefer and Chipman, Remote Sensing and Image Interpretation (Wiley)	CUGB Remote Sensing Geology (S5); MIT 12.421; Mines GEOL470; TH 40040913
	Sedimentology and Sedimentary Basins	3	Sediment transport, depositional environments and facies, diagenesis, basin types and subsidence, sediment-hosted resources.	Nichols, Sedimentology and Stratigraphy (Wiley-Blackwell); Allen and Allen, Basin Analysis (Wiley-Blackwell)	MIT 12.110A/B; Mines GEOL314/GEOL315
	GIS for Geoscientists	2	Spatial data models, coordinate systems, digitising geological maps, overlay and buffer analysis, DEMs, map production.	Bonham-Carter, Geographic Information Systems for Geoscientists (Pergamon); Longley et al., Geographic Information Science and Systems (Wiley)	Mines GEGN475 Applications of GIS; TH 20030212 Fundamentals of Geospatial Information Technology
Sem 6 20 cr total	**Mineral Exploration Methods**	4	Exploration stages, target generation, mineral-system models, integration of geological, geochemical and geophysical data, trenching and drilling grids, sampling.	Moon, Whateley and Evans (eds), Introduction to Mineral Exploration (Blackwell); Marjoribanks, Geological Methods in Mineral Exploration and Mining (Springer)	CUGB Mineral Prospecting and Exploration (S6); Mines GEGN403 (design); CUG-W Resource Exploration Eng. programme
	Exploration Geochemistry	3	Primary and secondary dispersion, stream-sediment, soil, rock and till surveys, sampling design, anomaly recognition and interpretation.	Rose, Hawkes and Webb, Geochemistry in Mineral Exploration (Academic Press); Levinson, Introduction to Exploration Geochemistry (Applied Publishing)	CUGB Exploration Geochemistry (S6)

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
	Exploration Drilling and Sampling	2	Diamond core, RC and percussion drilling, core logging, recovery and RQD, downhole surveys, sample preparation and QA/QC.	Marjoribanks, Geological Methods in Mineral Exploration and Mining (Springer); Heinz, Diamond Drill Handbook (Smith-Heinz)	Mines GEGN473 Site Investigation (drilling and sampling); CUG-W Exploration Technology and Engineering programme
	Mineral Resource Estimation and Geostatistics	3	Data analysis, compositing, variography, kriging, block models, resource classification, reconciliation.	Sinclair and Blackwell, Applied Mineral Inventory Estimation (Cambridge); Rossi and Deutsch, Mineral Resource Estimation (Springer)	Mines GEGN403 (reserve calculation part); MIT 12.012 (statistics)
	Introduction to Mining and Mineral Processing	2	Open-pit and underground mining methods, mine planning basics, comminution, flotation and other separation processes.	Hartman and Mutmansky, Introductory Mining Engineering (Wiley); Wills and Finch, Wills' Mineral Processing Technology (Butterworth-Heinemann)	CUGB Intro to Mining and Mineral Processing (S7); Mines GEOL311 Mining Geology
	Hydrogeology	2	Aquifers, Darcy's law, groundwater flow equations, well hydraulics and pumping tests, groundwater chemistry and contamination; mine and construction dewatering.	Fetter, Applied Hydrogeology (Waveland); Freeze and Cherry, Groundwater (Prentice Hall)	Mines GEGN466/466L Groundwater Engineering; TH 30040573 Hydrology (related)
	Metallogeny and Mineral Prospectivity	2	Metallogenic provinces and epochs, mineral systems approach, regional prospectivity analysis and target ranking.	Pirajno, Hydrothermal Processes and Mineral Systems (Springer); Carranza, Geochemical Anomaly and Mineral Prospectivity Mapping in GIS (Elsevier)	No direct equivalent in the five benchmark plans reviewed as a separate subject (part of CUGB Mineral Prospecting and Exploration)
	Mineral Economics and Project Evaluation	2	Commodity markets, cost estimation, cash-flow, NPV/IRR, sensitivity and risk, feasibility study stages.	Wellmer, Dalheimer and Wagner, Economic Evaluations in Exploration (Springer)	CUGB Mineral Resources Economics and Management (S6); Mines EBGN321 Engineering Economics
Summer 3 5 cr total	**Production Internship (5 weeks)**	5	Placement with an industry partner; daily log, supervisor assessment and internship report.	-	CUGB Production Field Trip (6 wks) + Mine-Base Training; TH 40030584 Construction Internship; TH 40041003 Production Internship
Sem 7 18 cr total	**Mineral Exploration Design (capstone)**	3	Team project: commodity and target selection, genetic model, programme and budget, drilling plan, preliminary resource and economics, report and presentation.	Moon, Whateley and Evans (eds), Introduction to Mineral Exploration (Blackwell)	Mines GEGN403 Mineral Exploration Design; CUGB Mineral Exploration Course Design (S6)
	Reporting Codes, Exploration Standards and Mineral Law	2	National exploration standards, CRIRSCO-type reporting (JORC, NI 43-101), competent-person concept, mineral rights and environmental law.	The JORC Code (Joint Ore Reserves Committee); national mineral resources law	CUGB Solid Mineral Exploration Standards + Environmental Law and Mineral Resources Law (S7)
	3D Geological Modelling	2	Implicit and explicit modelling, wireframes, drill-hole databases, block models, visualisation with mining software.	Mallet, Geomodeling (Oxford); Pyrcz and Deutsch, Geostatistical Reservoir Modeling (Oxford)	No direct equivalent in the five benchmark plans reviewed
	Coal and Industrial Minerals Geology	3	Coal formation, rank and quality, coal exploration; geology and specifications of industrial minerals and rocks.	Thomas, Coal Geology (Wiley-Blackwell); Kogel et al. (eds), Industrial Minerals and Rocks (SME)	CUG-W Resource Exploration Eng. (new-energy class); CUGB School of Energy Resources
	Environmental Geology and Mine Closure	2	Acid mine drainage, tailings and waste rock, environmental baselines, rehabilitation and closure planning.	Lottermoser, Mine Wastes: Characterization, Treatment and Environmental Impacts (Springer)	Mines Environmental Geosciences track (GEGN466/470); MIT 12.104
	Professional English and Technical Communication	2	Reading literature in the discipline, technical report and paper writing, presentations, international standards vocabulary.	Day and Gastel, How to Write and Publish a Scientific Paper (Cambridge)	CUGB Professional English for Resource Prospecting (S7); Mines writing-intensive (WI) capstones
	Elective (from any course of the faculty)	4	Chosen with the adviser from another course's list to broaden the specialisation.	-	TH category extension (6 cr); Mines free electives
Sem 8 10 cr total	**Graduation Design / Bachelor Thesis**	8	Independent project under supervision: literature review, data collection or design, analysis, thesis and oral defence.	-	TH Comprehensive Thesis Training (15 cr); CUGB BS Thesis (6 cr, 12 wks); MIT 12.TIP + 12.THU
	Frontiers Seminar of the Discipline	2	Guest lectures from industry and research institutes on new technologies, digitalisation and sustainability.	-	CUGB Frontiers of the Discipline (S7); MIT 12.00

2. Civil Engineering course

Responsible unit: Institute of Civil Engineering (civil building and prospecting, civil building and engineering divisions). Graduate profile: Civil engineers who can investigate ground conditions, design concrete and steel structures, foundations and underground works, and manage construction.

Course-specific credits: Sem 4 12, Summer 2 4, Sem 5 20, Sem 6 20, Summer 3 5, Sem 7 18, Sem 8 10 = 89. Semesters 1-3 and the common part of Sem 4 (8 cr): see section B. Total degree: 164 credits.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Sem 4 12 cr total	**Structural Analysis I**	4	Statically determinate and indeterminate beams, frames and trusses; influence lines; force and displacement methods; moment distribution.	Hibbeler, Structural Analysis (Pearson)	TH 20030044 Structural Mechanics (1) (Y3 autumn, 4 cr)
	Construction Materials (with laboratory)	3	Aggregates, cement and concrete, mix design, steel, timber, masonry, asphalt; standard tests.	Mamlouk and Zaniewski, Materials for Civil and Construction Engineers (Pearson); Mehta and Monteiro, Concrete (McGraw-Hill)	TH 40030352 Building Materials + 40030361 Building Materials Lab
	Soil Mechanics (with laboratory)	3	Soil classification, compaction, permeability and seepage, effective stress, consolidation, shear strength, lateral earth pressure.	Das and Sobhan, Principles of Geotechnical Engineering (Cengage); Knappett and Craig, Craig's Soil Mechanics (CRC)	TH 30040513 Soil Mechanics; Mines CEEN312/312L Soil Mechanics
	Engineering Surveying	2	Levelling, angle and distance measurement, total station, traversing, GNSS, topographic surveys, setting-out of works.	Ghilani and Wolf, Elementary Surveying: An Introduction to Geomatics (Pearson)	TH 30030611 Surveying + 40030402 Surveying Practice; CUGB Surveying A (S4)
Summer 2 4 cr total	**Surveying and Engineering Geology Field Practice (4 weeks)**	4	2 weeks surveying camp (control network, topographic map); 2 weeks engineering-geological mapping of a construction site.	Ghilani and Wolf, Elementary Surveying (Pearson)	TH 40030402 Surveying Practice (2 wks); TH 30040551 Geology Practice
Sem 5 20 cr total	**Structural Analysis II (Matrix and Finite Element Methods)**	3	Stiffness method for trusses and frames, introduction to finite elements, use of structural software.	McGuire, Gallagher and Ziemian, Matrix Structural Analysis (Wiley); Logan, A First Course in the Finite Element Method (Cengage)	TH 40030322 Structural Matrix Analysis; TH 20030102 Elasticity and Finite Elements
	Reinforced Concrete Design I	4	Material behaviour, flexure, shear, bond and anchorage, columns, serviceability; design to the national code.	Wight, Reinforced Concrete: Mechanics and Design (Pearson); McCormac and Brown, Design of Reinforced Concrete (Wiley)	TH 20030153 Concrete Structures 1; TH 30030583 Reinforced Concrete
	Steel Structures	3	Tension and compression members, beams, beam-columns, connections (bolted, welded), stability.	Segui, Steel Design (Cengage)	TH 30030113 Steel Structures (1)
	Foundation Engineering	3	Site investigation, bearing capacity, settlement, shallow foundations, piles, retaining walls, ground improvement.	Das and Sivakugan, Principles of Foundation Engineering (Cengage)	TH 30040162 Foundation Engineering
	Engineering Geology and Site Investigation	3	Geological factors in engineering, ground investigation (drilling, sampling, in-situ tests), geotechnical reports, geohazards.	Gonzalez de Vallejo and Ferrer, Geological Engineering (CRC); Clayton, Matthews and Simons, Site Investigation (Blackwell)	TH 40040152 Engineering Geology; Mines GEGN468 Engineering Geology and Geotechnics, GEGN473 Site Investigation; CUG-W Geological Engineering programme
	Engineering Hydraulics and Hydrology	2	Hydrological cycle, rainfall-runoff, design floods, open-channel flow, culverts, drainage and stormwater design.	Mays, Water Resources Engineering (Wiley); Chow, Open-Channel Hydraulics (McGraw-Hill)	TH 30040573 Hydrology, 20040143 River Hydraulics
	Building Architecture and Design	2	Functional planning, building elements, envelopes, stairs, roofs, fire and accessibility requirements, architectural drawings.	Ching, Building Construction Illustrated (Wiley)	TH 30030522 Building Architecture; TH 40031102 Fundamentals of Architecture
Sem 6 20 cr total	**Reinforced Concrete Design II (Buildings)**	3	Slabs, frames, shear walls, foundations of multi-storey buildings, prestressed concrete introduction.	Wight, Reinforced Concrete: Mechanics and Design (Pearson)	TH 20030162 Concrete Structures (2)
	Structural Dynamics and Earthquake Engineering	3	Single and multi-degree-of-freedom systems, response spectra, seismic design principles, ductile detailing.	Chopra, Dynamics of Structures (Pearson)	TH 30030601 Earthquake Engineering and Dynamics; TH 30040042 Structural Dynamics; MIT 12.210 (seismology basis)
	Rock Mechanics	3	Stress and strain in rock, strength criteria (Mohr-Coulomb, Hoek-Brown), discontinuities, rock mass classification (RMR, Q, GSI), laboratory and in-situ tests.	Hudson and Harrison, Engineering Rock Mechanics (Elsevier); Goodman, Introduction to Rock Mechanics (Wiley)	TH 30040142 Rock Mechanics; Mines MNGN321 Intro to Rock Mechanics
	Construction Technology and Management	3	Earthworks, concrete and formwork, construction equipment; planning, CPM scheduling, cost and quality control, site safety.	Halpin, Senior and Lucko, Construction Management (Wiley); Nunnally, Construction Methods and Management (Pearson)	TH 30910032 Construction Technology, 30910042 Construction Organisation, 30910052 Project Management I

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
	Road and Bridge Engineering	3	Highway geometric design, pavement design, bridge types, loads and design of short-span bridges.	Garber and Hoel, Traffic and Highway Engineering (Cengage); Barker and Puckett, Design of Highway Bridges (Wiley)	TH 40030492 Road Engineering; TH 40030782 Bridge Engineering
	Engineering Economics	2	Time value of money, cash-flow analysis, NPV, IRR, benefit-cost, depreciation and taxes, replacement decisions.	Newnan, Eschenbach and Lavelle, Engineering Economic Analysis (Oxford)	TH 30030352 Engineering Economics; Mines EBGN321 Engineering Economics
	BIM and Computer-Aided Structural Design	3	Building information modelling, structural design software, drawing production, clash detection.	Sacks, Eastman, Lee and Teicholz, BIM Handbook (Wiley)	TH 30030323 Civil Engineering CAD; TH 40031062 Intelligent Construction
Summer 3 5 cr total	**Production Internship (5 weeks)**	5	Placement with an industry partner; daily log, supervisor assessment and internship report.	-	CUGB Production Field Trip (6 wks) + Mine-Base Training; TH 40030584 Construction Internship; TH 40041003 Production Internship
Sem 7 18 cr total	**Integrated Civil Engineering Design (capstone)**	4	Team design of a building or civil structure from site investigation to structural design, drawings, cost estimate and construction plan.	National design codes; Ching, Building Construction Illustrated (Wiley)	TH 40031184 Integrated Course Design (Civil); Mines GEGN469 Engineering Geology Design
	Slope Stability and Geohazard Mitigation	3	Limit-equilibrium analysis, rock slope kinematics, landslides, debris flows, stabilisation measures and monitoring.	Duncan, Wright and Brandon, Soil Strength and Slope Stability (Wiley); Wyllie, Rock Slope Engineering (CRC)	Mines GEGN468/469; TH 40040782 Engineering Disaster Science
	Building Services and Green Building	2	Water supply, drainage, HVAC and electrical services; energy efficiency, green-building rating systems.	Kibert, Sustainable Construction: Green Building Design and Delivery (Wiley)	TH 00990021 Green Building and Sustainable Development
	Construction Contracts, Codes and Law	2	Contract types, tendering, claims, building regulations, professional liability.	Fisk and Reynolds, Construction Project Administration (Pearson)	TH 30030482 Engineering Contract Management; TH 40031141 Construction HSE Management
	Tunnelling and Underground Space Engineering	3	Underground space planning, excavation methods (drill-and-blast, TBM, NATM), support design, monitoring.	Chapman, Metje and Stark, Introduction to Tunnel Construction (CRC); Hoek and Brown, Underground Excavations in Rock (CRC)	TH 40030152 Underground Structures, 40040042 Underground Cavern Eng.; CUG-W Urban Underground Space Eng. programme
	Professional English and Technical Communication	2	Reading literature in the discipline, technical report and paper writing, presentations, international standards vocabulary.	Day and Gastel, How to Write and Publish a Scientific Paper (Cambridge)	CUGB Professional English for Resource Prospecting (S7); Mines writing-intensive (WI) capstones
	Elective (from any course of the faculty)	2	Chosen with the adviser from another course's list to broaden the specialisation.	-	TH category extension (6 cr); Mines free electives
Sem 8 10 cr total	**Graduation Design / Bachelor Thesis**	8	Independent project under supervision: literature review, data collection or design, analysis, thesis and oral defence.	-	TH Comprehensive Thesis Training (15 cr); CUGB BS Thesis (6 cr, 12 wks); MIT 12.TIP + 12.THU
	Frontiers Seminar of the Discipline	2	Guest lectures from industry and research institutes on new technologies, digitalisation and sustainability.	-	CUGB Frontiers of the Discipline (S7); MIT 12.00

3. Geophysical Exploration course

Responsible unit: Institute of Geophysical Prospecting (electromagnetic, magnetic, prospecting-equipment divisions). Graduate profile: Geophysicists who can design, acquire, process and interpret gravity, magnetic, electrical, EM and seismic surveys, and develop field instruments.

Course-specific credits: Sem 4 12, Summer 2 4, Sem 5 20, Sem 6 20, Summer 3 5, Sem 7 18, Sem 8 10 = 89. Semesters 1-3 and the common part of Sem 4 (8 cr): see section B. Total degree: 164 credits.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Sem 4 12 cr total	**Fundamentals of Geophysics**	3	Earth's gravity and figure, geomagnetism, seismology and Earth structure, heat flow, geochronology; physical properties of rocks.	Lowrie and Fichtner, Fundamentals of Geophysics (Cambridge)	MIT 12.002 Intro to Geophysics and Planetary Science, 12.201 Essentials of Global Geophysics
	Signals, Systems and Digital Signal Processing	3	Continuous and discrete signals, convolution, Fourier and z-transforms, sampling, filters, spectral analysis of geophysical time series.	Oppenheim, Willsky and Nawab, Signals and Systems (Pearson); Gubbins, Time Series Analysis and Inverse Theory for Geophysicists (Cambridge)	MIT 12.012 Statistics, Regression, Signal Processing
	Electromagnetic Field Theory	3	Vector analysis, electrostatics, magnetostatics, Maxwell's equations, wave propagation and diffusion in conductive media.	Cheng, Field and Wave Electromagnetics (Pearson); Griffiths, Introduction to Electrodynamics (Cambridge)	No direct equivalent in the five benchmark plans reviewed (physics-department subject)
	Petrology and Structural Geology for Geophysicists	3	Rock types and their physical properties; folds, faults and structural settings relevant to anomaly interpretation.	Klein and Philpotts, Earth Materials (Cambridge); Fossen, Structural Geology (Cambridge)	Mines GEOL308 Intro Applied Structural Geology; MIT 12.113
Summer 2 4 cr total	**Geological and Geophysical Field Practice (4 weeks)**	4	2 weeks geological mapping, 2 weeks ground magnetic, gravity and resistivity surveys over a known target.	Milsom and Eriksen, Field Geophysics (Wiley)	MIT 12.211 Field Geophysics / 12.214; Mines GEGN316
Sem 5 20 cr total	**Gravity and Magnetic Prospecting**	4	Potential theory, instruments, data reduction, anomalies of simple bodies, filtering, derivatives, Euler deconvolution, 2D/3D modelling.	Blakely, Potential Theory in Gravity and Magnetic Applications (Cambridge); Hinze, von Frese and Saad, Gravity and Magnetic Exploration (Cambridge)	CUG-W Exploration Geophysics programme; CUGB Dept. of Geophysics; MIT 12.201
	Electrical Prospecting (Resistivity, IP, SP)	4	DC resistivity theory, arrays, vertical sounding and electrical resistivity tomography, induced polarisation, self-potential; interpretation.	Telford, Geldart and Sheriff, Applied Geophysics (Cambridge); Reynolds, An Introduction to Applied and Environmental Geophysics (Wiley-Blackwell)	CUGB Exploration Geophysics Methods (S5); CUG-W Exploration Geophysics programme
	Seismic Exploration	4	Elastic waves, reflection and refraction, acquisition geometry, sources and receivers, velocity analysis basics.	Sheriff and Geldart, Exploration Seismology (Cambridge)	MIT 12.210 Introduction to Seismology; Mines GEGN439 (seismic interpretation)
	Petrophysics and Rock Physics	2	Density, magnetic susceptibility, resistivity, chargeability, elastic properties of rocks; laboratory measurement; property-based target models.	Schon, Physical Properties of Rocks (Elsevier); Mavko, Mukerji and Dvorkin, The Rock Physics Handbook (Cambridge)	MIT 12.202 Flow, Deformation and Fracture; MIT Earth Resources Laboratory research
	Geophysical Inverse Theory	3	Forward and inverse problems, least squares, regularisation, non-linear and stochastic inversion, resolution and uncertainty.	Menke, Geophysical Data Analysis: Discrete Inverse Theory (Academic Press); Aster, Borchers and Thurber, Parameter Estimation and Inverse Problems (Elsevier)	MIT 12.012 (part); MIT graduate geophysics
	Geophysical Instrumentation	3	Magnetometers, gravimeters, resistivity/IP receivers, EM transmitters/receivers, seismographs; noise, calibration, GNSS timing.	Milsom and Eriksen, Field Geophysics (Wiley); Horowitz and Hill, The Art of Electronics (Cambridge)	CUGB Dept. of Measurement and Control Technology; CUG-W Mech. Eng. and Electronic Information (intelligent geo-instruments)
Sem 6 20 cr total	**Electromagnetic Prospecting (TEM, MT, CSAMT, GPR)**	4	Frequency- and time-domain EM, airborne EM, magnetotellurics and CSAMT, ground-penetrating radar; 1D/2D inversion.	Nabighian (ed.), Electromagnetic Methods in Applied Geophysics (SEG); Simpson and Bahr, Practical Magnetotellurics (Cambridge)	CUG-W Exploration Geophysics programme; CUGB Dept. of Geophysics
	Seismic Data Processing	4	Geometry, statics, deconvolution, velocity analysis, NMO and stacking, migration, noise attenuation.	Yilmaz, Seismic Data Analysis (SEG)	Mines Geophysical Engineering programme; MIT Earth Resources Laboratory
	Well Logging and Formation Evaluation	3	Electrical, nuclear, acoustic and imaging logs; porosity, saturation and lithology interpretation; logs in mineral boreholes.	Ellis and Singer, Well Logging for Earth Scientists (Springer); Rider and Kennedy, The Geological Interpretation of Well Logs (Rider-French)	Mines GEGN438 Petroleum Geology (log lab)
	Ore Deposit Geology	2	Ore-forming processes: magmatic, hydrothermal, sedimentary, supergene; deposit models (porphyry, VMS, epithermal, orogenic gold, skarn, SEDEX, IOCG, REE); ore mineral assemblages.	Robb, Introduction to Ore-Forming Processes (Wiley-Blackwell); Pohl, Economic Geology: Principles and Practice (Wiley-Blackwell)	CUGB Ore Deposit Geology (S5, 4 cr); Mines GEGN401 Mineral Deposits; CUG-W Ore Deposit Geology Field Training

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
	Geophysical Equipment Design	3	Sensor design, analogue front ends, embedded systems, low-noise power, data logging and wireless telemetry; a small instrument project.	Fraden, Handbook of Modern Sensors (Springer); Horowitz and Hill, The Art of Electronics (Cambridge)	CUGB Dept. of Measurement and Control Technology; CUG-W intelligent geological equipment
	Airborne and UAV Geophysics	2	Airborne magnetics, radiometrics and EM; UAV platforms; survey planning, compensation, levelling and gridding.	Reeves, Aeromagnetic Surveys: Principles, Practice and Interpretation (Geosoft); Dentith and Mudge (Cambridge)	No direct equivalent in the five benchmark plans reviewed
	Numerical Modelling in Geophysics	2	Finite-difference and finite-element forward modelling for potential-field, EM and seismic problems.	Igel, Computational Seismology: A Practical Introduction (Oxford)	MIT 12.010 Computational Methods
Summer 3 5 cr total	**Production Internship (5 weeks)**	5	Placement with an industry partner; daily log, supervisor assessment and internship report.	-	CUGB Production Field Trip (6 wks) + Mine-Base Training; TH 40030584 Construction Internship; TH 40041003 Production Internship
Sem 7 18 cr total	**Seismic Interpretation**	3	Structural and stratigraphic interpretation of 2D/3D seismic, attributes, depth conversion, well ties.	Bacon, Simm and Redshaw, 3-D Seismic Interpretation (Cambridge); Brown, Interpretation of Three-Dimensional Seismic Data (AAPG/SEG)	Mines GEGN439 Petroleum Exploration Design (seismic in Petrel)
	Integrated Geophysical Interpretation and Survey Design (capstone)	3	Team project: integrate several methods over a real dataset, design a survey, joint interpretation and report.	Dentith and Mudge, Geophysics for the Mineral Exploration Geoscientist (Cambridge)	Mines GEGN439 Petroleum Exploration Design; MIT 12.211 Field Geophysics
	Engineering and Environmental Geophysics	3	Shallow seismic, ERT, GPR and EM for site investigation, groundwater, cavities, contamination and archaeology.	Reynolds, An Introduction to Applied and Environmental Geophysics (Wiley-Blackwell)	MIT 12.214 Essentials of Field Geophysics
	Radiometric and Nuclear Prospecting	2	Radioactivity, gamma-ray spectrometry, radon methods, airborne radiometrics, uranium exploration, radiation safety.	IAEA, Guidelines for Radioelement Mapping Using Gamma Ray Spectrometry Data (IAEA-TECDOC-1363)	No direct equivalent in the five benchmark plans reviewed
	Geophysical Equipment Development Project	3	Design, build and field-test a prototype instrument or software module in teams.	Fraden, Handbook of Modern Sensors (Springer)	CUG-W Engineering Research Centre of Rock-Soil Drilling and Equipment (research link)
	Professional English and Technical Communication	2	Reading literature in the discipline, technical report and paper writing, presentations, international standards vocabulary.	Day and Gastel, How to Write and Publish a Scientific Paper (Cambridge)	CUGB Professional English for Resource Prospecting (S7); Mines writing-intensive (WI) capstones
	Elective (from any course of the faculty)	2	Chosen with the adviser from another course's list to broaden the specialisation.	-	TH category extension (6 cr); Mines free electives
Sem 8 10 cr total	**Graduation Design / Bachelor Thesis**	8	Independent project under supervision: literature review, data collection or design, analysis, thesis and oral defence.	-	TH Comprehensive Thesis Training (15 cr); CUGB BS Thesis (6 cr, 12 wks); MIT 12.TIP + 12.THU
	Frontiers Seminar of the Discipline	2	Guest lectures from industry and research institutes on new technologies, digitalisation and sustainability.	-	CUGB Frontiers of the Discipline (S7); MIT 12.00

4. Geochemical Exploration course

Responsible unit: Institute of Resource Exploration (geochemical prospecting division). Graduate profile: Exploration and environmental geochemists who can plan surveys, run and control laboratory analyses, process multivariate data and interpret anomalies.

Course-specific credits: Sem 4 12, Summer 2 4, Sem 5 20, Sem 6 20, Summer 3 5, Sem 7 18, Sem 8 10 = 89. Semesters 1-3 and the common part of Sem 4 (8 cr): see section B. Total degree: 164 credits.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Sem 4 12 cr total	**Petrology and Optical Petrography**	4	Optical mineralogy; igneous, metamorphic and sedimentary petrology; thin-section description; phase diagrams; rocks as records of tectonic setting.	Philpotts and Ague, Principles of Igneous and Metamorphic Petrology (Cambridge); Nesse, Introduction to Optical Mineralogy (Oxford)	CUGB Petrology (S4, 5.5 cr); Mines GEGN307 Petrology; MIT 12.108
	Structural Geology	4	Stress and strain, faults, folds, foliations, shear zones, stereographic projection, structural analysis of maps and cross-sections; structural control of ore and traps.	Fossen, Structural Geology (Cambridge); Lisle, Geological Structures and Maps (Elsevier)	CUGB Structural Geology (S4); Mines GEOL309; MIT 12.113
	Analytical Chemistry (Quantitative and Instrumental)	4	Gravimetric and volumetric analysis, spectroscopy, chromatography, electrochemistry, calibration and error.	Harris, Quantitative Chemical Analysis (W.H. Freeman); Skoog, Holler and Crouch, Principles of Instrumental Analysis (Cengage)	CUGB Physical Chemistry C (S3) (related); Mines CHGN122
Summer 2 4 cr total	**Geological Mapping Field School (4 weeks)**	4	Systematic mapping of a sedimentary-igneous-metamorphic terrain; measured sections; structural data; geological map, cross-sections and field report.	Lisle, Brabham and Barnes, Basic Geological Mapping (Wiley-Blackwell)	CUGB Zhoukoudian Mapping Field School (6 wks); Mines GEGN316 Field Geology (5 wks); MIT 12.115
Sem 5 20 cr total	**Geochemistry**	4	Element distribution and cosmochemistry, crystal chemistry, thermodynamics of geologic systems, aqueous geochemistry, trace elements, isotope basics, organic geochemistry introduction.	White, Geochemistry (Wiley-Blackwell); Misra, Introduction to Geochemistry: Principles and Applications (Wiley-Blackwell)	CUGB Geochemistry incl. organic (S5); Mines CHGN406 Intro to Geochemistry; MIT 12.104
	Ore Deposit Geology	4	Ore-forming processes: magmatic, hydrothermal, sedimentary, supergene; deposit models (porphyry, VMS, epithermal, orogenic gold, skarn, SEDEX, IOCG, REE); ore mineral assemblages.	Robb, Introduction to Ore-Forming Processes (Wiley-Blackwell); Pohl, Economic Geology: Principles and Practice (Wiley-Blackwell)	CUGB Ore Deposit Geology (S5, 4 cr); Mines GEGN401 Mineral Deposits; CUG-W Ore Deposit Geology Field Training
	Instrumental Geoanalysis	3	Sample preparation, XRF, ICP-OES/MS, AAS, LA-ICP-MS, electron microprobe and SEM; detection limits and standards.	Potts, A Handbook of Silicate Rock Analysis (Blackie); Reed, Electron Microprobe Analysis and Scanning Electron Microscopy in Geology (Cambridge)	Mines GEOL321 (XRD, SEM labs); MIT 12.335 Experimental Atmospheric Chemistry (lab methods)
	Isotope Geochemistry and Geochronology	3	Radiogenic isotopes (Rb-Sr, Sm-Nd, U-Pb), stable isotopes (O, H, C, S), dating methods and ore-fluid sources.	Faure and Mensing, Isotopes: Principles and Applications (Wiley); Dickin, Radiogenic Isotope Geology (Cambridge)	MIT 12.104 (part); MIT graduate geochemistry
	Aqueous Geochemistry and Geochemical Thermodynamics	3	Activity, equilibrium, mineral solubility, redox, Eh-pH diagrams, speciation modelling (PHREEQC).	Langmuir, Aqueous Environmental Geochemistry (Prentice Hall); Anderson, Thermodynamics of Natural Systems (Cambridge)	MIT 12.104 Geochemistry of Natural Waters; Mines GEGN330 Geoscientists Thermodynamics
	Applied Geophysics for Exploration	3	Gravity, magnetic, electrical, electromagnetic, radiometric and seismic methods; survey design and interpretation for mineral exploration.	Dentith and Mudge, Geophysics for the Mineral Exploration Geoscientist (Cambridge); Kearey, Brooks and Hill, An Introduction to Geophysical Exploration (Wiley-Blackwell)	CUGB Exploration Geophysics Methods and Applications (S5); MIT 12.002, 12.211; Mines GEGN439 (integration)
Sem 6 20 cr total	**Exploration Geochemistry I (Survey Methods)**	4	Dispersion processes, stream-sediment, soil, rock, till and lake-sediment surveys, sampling density, field and laboratory protocols.	Rose, Hawkes and Webb, Geochemistry in Mineral Exploration (Academic Press); Levinson, Introduction to Exploration Geochemistry (Applied Publishing)	CUGB Exploration Geochemistry (S6)
	Geochemical Data Processing and Multivariate Statistics	3	Data distributions, background and threshold, compositional data (log-ratio), PCA, factor and cluster analysis, geochemical mapping in R/Python.	Reimann, Filzmoser, Garrett and Dutter, Statistical Data Analysis Explained (Wiley); Davis, Statistics and Data Analysis in Geology (Wiley)	Mines GEGN432 Geological Data Management; MIT 12.012
	Biogeochemical and Gas Methods	2	Plant and humus sampling, geobotany, soil-gas and Hg/radon surveys, partial and selective leaches.	Dunn, Biogeochemistry in Mineral Exploration (Elsevier)	MIT 12.107 Geobiology (related)
	Hydrogeochemistry	2	Groundwater chemistry evolution, water-rock interaction, hydrogeochemical exploration, water quality.	Appelo and Postma, Geochemistry, Groundwater and Pollution (CRC)	MIT 12.104; Mines GEGN466

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
	Environmental Geochemistry	3	Potentially toxic elements, soil and sediment contamination, geochemical baselines, risk assessment.	Siegel, Environmental Geochemistry of Potentially Toxic Metals (Springer); Lottermoser, Mine Wastes (Springer)	MIT 12.104; Mines Environmental Geosciences track
	Organic Geochemistry	2	Kerogen types, maturation, biomarkers, source-rock evaluation, hydrocarbon geochemistry for petroleum exploration.	Tissot and Welte, Petroleum Formation and Occurrence (Springer); Killops and Killops, Introduction to Organic Geochemistry (Blackwell)	CUGB Geochemistry incl. organic geochemistry (S5)
	Mineral Exploration Methods	2	Exploration stages, target generation, mineral-system models, integration of geological, geochemical and geophysical data, trenching and drilling grids, sampling.	Moon, Whateley and Evans (eds), Introduction to Mineral Exploration (Blackwell); Marjoribanks, Geological Methods in Mineral Exploration and Mining (Springer)	CUGB Mineral Prospecting and Exploration (S6); Mines GEGN403 (design); CUG-W Resource Exploration Eng. programme
	GIS for Geoscientists	2	Spatial data models, coordinate systems, digitising geological maps, overlay and buffer analysis, DEMs, map production.	Bonham-Carter, Geographic Information Systems for Geoscientists (Pergamon); Longley et al., Geographic Information Science and Systems (Wiley)	Mines GEGN475 Applications of GIS; TH 20030212 Fundamentals of Geospatial Information Technology
Summer 3 5 cr total	**Production Internship (5 weeks)**	5	Placement with an industry partner; daily log, supervisor assessment and internship report.	-	CUGB Production Field Trip (6 wks) + Mine-Base Training; TH 40030584 Construction Internship; TH 40041003 Production Internship
Sem 7 18 cr total	**Exploration Geochemistry II (Anomaly Interpretation and Regional Mapping)**	3	Regional geochemical mapping, anomaly ranking, integration with geology and geophysics, case histories by deposit type.	Govett (ed.), Handbook of Exploration Geochemistry (Elsevier); Carranza, Geochemical Anomaly and Mineral Prospectivity Mapping in GIS (Elsevier)	CUGB Exploration Geochemistry (S6)
	Geochemical Survey Design (capstone)	3	Team design of a geochemical programme: objectives, media, sampling plan, QA/QC, budget, interpretation of supplied data, report.	Rose, Hawkes and Webb, Geochemistry in Mineral Exploration (Academic Press)	CUGB Mineral Exploration Course Design (S6); Mines GEGN403
	QA/QC and Laboratory Management	2	Standards, blanks and duplicates, control charts, accreditation (ISO/IEC 17025), data management.	Abzalov, Applied Mining Geology (Springer)	No direct equivalent in the five benchmark plans reviewed
	Mineral Resource Estimation and Geostatistics	2	Data analysis, compositing, variography, kriging, block models, resource classification, reconciliation.	Sinclair and Blackwell, Applied Mineral Inventory Estimation (Cambridge); Rossi and Deutsch, Mineral Resource Estimation (Springer)	Mines GEGN403 (reserve calculation part); MIT 12.012 (statistics)
	Geochemistry of the Mining Environment and Remediation	3	Acid drainage prediction tests, metal mobility, tailings geochemistry, passive and active treatment, phytoremediation.	Lottermoser, Mine Wastes: Characterization, Treatment and Environmental Impacts (Springer)	Mines GEGN470 Ground-water Engineering Design (water quality)
	Professional English and Technical Communication	2	Reading literature in the discipline, technical report and paper writing, presentations, international standards vocabulary.	Day and Gastel, How to Write and Publish a Scientific Paper (Cambridge)	CUGB Professional English for Resource Prospecting (S7); Mines writing-intensive (WI) capstones
	Elective (from any course of the faculty)	3	Chosen with the adviser from another course's list to broaden the specialisation.	-	TH category extension (6 cr); Mines free electives
Sem 8 10 cr total	**Graduation Design / Bachelor Thesis**	8	Independent project under supervision: literature review, data collection or design, analysis, thesis and oral defence.	-	TH Comprehensive Thesis Training (15 cr); CUGB BS Thesis (6 cr, 12 wks); MIT 12.TIP + 12.THU
	Frontiers Seminar of the Discipline	2	Guest lectures from industry and research institutes on new technologies, digitalisation and sustainability.	-	CUGB Frontiers of the Discipline (S7); MIT 12.00

5. Drill Engineering course

Responsible unit: Independent Drill Engineering Division. Graduate profile: Drilling engineers for mineral-exploration, water-well, geotechnical, geothermal and petroleum drilling: method selection, rig and bit design, fluids, well design and safety.

Course-specific credits: Sem 4 12, Summer 2 4, Sem 5 20, Sem 6 20, Summer 3 5, Sem 7 18, Sem 8 10 = 89. Semesters 1-3 and the common part of Sem 4 (8 cr): see section B. Total degree: 164 credits.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Sem 4 12 cr total	**Rock Mechanics and Rock Breaking**	3	Rock strength and drillability, fracture mechanics, mechanisms of cutting, crushing and percussive breakage, rock abrasivity.	Jaeger, Cook and Zimmerman, Fundamentals of Rock Mechanics (Wiley-Blackwell); Hudson and Harrison, Engineering Rock Mechanics (Elsevier)	TH 30040142 Rock Mechanics; Mines MNGN321
	Machine Elements and Mechanical Design	4	Stress and fatigue, shafts, bearings, gears, couplings, threads, springs, design of drill-rig components.	Budynas and Nisbett, Shigley's Mechanical Engineering Design (McGraw-Hill)	CUGB Dept. of Mechanical Design and Automation
	Hydraulic and Pneumatic Drives	3	Pumps, motors, cylinders, valves, circuits, compressors and air systems for drilling rigs.	Esposito, Fluid Power with Applications (Pearson)	CUG-W School of Mechanical Engineering and Electronic Information
	Geology for Drilling Engineers	2	Rock types and drillability, structures, weathering, groundwater and hazardous ground conditions; core description.	Klein and Philpotts, Earth Materials (Cambridge); Fossen, Structural Geology (Cambridge)	Mines GEOL310 Earth Materials (for mining engineers)
Summer 2 4 cr total	**Drilling Equipment and Geology Field Practice (4 weeks)**	4	2 weeks geological mapping, 2 weeks on a teaching drill site: rig components, operation, core handling.	Australian Drilling Industry Training Committee, Drilling: The Manual of Methods, Applications and Management (CRC)	CUG-W Exploration Technology and Engineering practice; Mines Edgar Experimental Mine
Sem 5 20 cr total	**Drilling Engineering I (Methods)**	4	Rotary, diamond core, reverse-circulation, percussion, cable-tool and auger drilling; selection of method; drilling parameters and rate of penetration.	Australian Drilling Industry Training Committee, Drilling: The Manual of Methods, Applications and Management (CRC)	CUG-W Exploration Technology and Engineering programme
	Drilling Fluids and Hydraulics	3	Functions and types of drilling fluids, rheology, solids control, hole cleaning, pressure losses, pump selection.	Caenn, Darley and Gray, Composition and Properties of Drilling and Completion Fluids (Gulf)	CUG-W Exploration Technology and Engineering programme
	Drilling Machinery and Equipment	4	Rig types, hoisting, rotation and circulation systems, power units, drill string, maintenance and reliability.	Bourgoyne, Millheim, Chenevert and Young, Applied Drilling Engineering (SPE)	CUG-W Mech. Eng. (geological drilling equipment); CUGB Cooperation Base for Deep Drilling Equipment
	Drill Bits and Diamond Tools	3	Roller-cone, PDC, impregnated and surface-set diamond bits, powder metallurgy of matrices, bit selection and wear analysis.	Konstanty, Powder Metallurgy Diamond Tools (Elsevier); Heinz, Diamond Drill Handbook (Smith-Heinz)	CUG-W Exploration Technology and Engineering programme
	Engineering Thermodynamics and Heat Transfer	2	Laws of thermodynamics, cycles of engines and compressors, conduction, convection, downhole temperature.	Cengel and Boles, Thermodynamics: An Engineering Approach (McGraw-Hill)	Mines GEGN330 Geoscientists Thermodynamics
	Electrical Drives and Automation	2	Motors and variable-frequency drives, PLCs, sensors and control loops for rig automation.	Bolton, Mechatronics (Pearson)	CUG-W School of Automation
	Geotechnical Engineering Basics	2	Soil classification, effective stress, consolidation and shear strength, bearing capacity basics.	Das and Sobhan, Principles of Geotechnical Engineering (Cengage)	TH 30040513 Soil Mechanics; Mines CEEN312
Sem 6 20 cr total	**Drilling Engineering II (Core Drilling for Exploration)**	3	Wireline coring, core recovery in difficult ground, borehole deviation control and surveying, deep core drilling.	Heinz, Diamond Drill Handbook (Smith-Heinz); Marjoribanks, Geological Methods in Mineral Exploration and Mining (Springer)	CUGB Laboratory of Deep Geological Drilling Technology (research link); Mines GEGN473
	Directional and Horizontal Drilling	3	Well-path planning, deflection tools, mud motors, rotary steerable systems, surveying and anti-collision.	Inglis, Directional Drilling (Springer); Mitchell and Miska, Fundamentals of Drilling Engineering (SPE)	Mines Petroleum Engineering programme
	Well Design: Casing, Cementing and Completion	3	Casing design, cementing operations and cement chemistry, completion types, well integrity.	Nelson and Guillot (eds), Well Cementing (Schlumberger); Rabia, Well Engineering and Construction (Entrac)	Mines Petroleum Engineering programme
	Water Well Drilling and Groundwater	3	Hydrogeology for well siting, well design, screens and gravel pack, development, pumping tests, maintenance.	Driscoll, Groundwater and Wells (Johnson Screens); Misstear, Banks and Clark, Water Wells and Boreholes (Wiley)	Mines GEGN466 Groundwater Engineering, GEGN470 Ground-water Engineering Design
	Geotechnical Drilling, In-situ Testing and Piling	3	Investigation boreholes, SPT, CPT, pressuremeter, packer tests; bored piles, anchors, grouting and jet grouting.	Clayton, Matthews and Simons, Site Investigation (Blackwell); Tomlinson and Woodward, Pile Design and Construction Practice (CRC)	Mines GEGN473 Geological Engineering Site Investigation; TH 30040162 Foundation Engineering

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
	Drilling Instrumentation and Monitoring (MWD/LWD)	3	Surface data acquisition, downhole sensors, measurement- and logging-while-drilling, telemetry, real-time interpretation.	Mitchell and Miska, Fundamentals of Drilling Engineering (SPE); Ellis and Singer, Well Logging for Earth Scientists (Springer)	Mines Petroleum Engineering programme
	Drilling Project Management and Cost Engineering	2	Tendering, contract types, cost per metre, scheduling, logistics, crew management, HSE plans.	Halpin, Senior and Lucko, Construction Management (Wiley)	TH 30910052 Engineering Project Management I
Summer 3 5 cr total	**Production Internship (5 weeks)**	5	Placement with an industry partner; daily log, supervisor assessment and internship report.	-	CUGB Production Field Trip (6 wks) + Mine-Base Training; TH 40030584 Construction Internship; TH 40041003 Production Internship
Sem 7 18 cr total	**Drilling Accidents, Well Control and Safety**	3	Stuck pipe, lost circulation, fishing operations, kicks and blowout prevention, H2S, rig safety management.	Grace, Blowout and Well Control Handbook (Gulf)	Mines Petroleum Engineering programme
	Drilling Project Design (capstone)	3	Team design of a complete drilling programme (exploration core hole, water well or geothermal well): rig, bits, fluids, casing, budget.	Bourgoyne et al., Applied Drilling Engineering (SPE)	Mines GEGN470 (drilling programme design); Mines GEGN439
	Trenchless Technology and Mechanised Tunnelling	3	Horizontal directional drilling, microtunnelling, pipe jacking, raise boring, TBMs.	Najafi, Trenchless Technology: Planning, Equipment, and Methods (McGraw-Hill)	TH 40040042 Underground Cavern Engineering; CUG-W Urban Underground Space Eng.
	Deep, Scientific and Geothermal Drilling	2	High-temperature drilling, scientific continental drilling, geothermal well design, special bits and fluids.	Finger and Blankenship, Handbook of Best Practices for Geothermal Drilling (Sandia National Laboratories)	CUGB Laboratory of Deep Geological Drilling Technology
	Drilling Automation and Intelligent Drilling	2	Automated rigs, drilling optimisation, data analytics and machine learning on drilling data, digital twins.	Mitchell and Miska, Fundamentals of Drilling Engineering (SPE)	No direct equivalent in the five benchmark plans reviewed
	Professional English and Technical Communication	2	Reading literature in the discipline, technical report and paper writing, presentations, international standards vocabulary.	Day and Gastel, How to Write and Publish a Scientific Paper (Cambridge)	CUGB Professional English for Resource Prospecting (S7); Mines writing-intensive (WI) capstones
	Elective (from any course of the faculty)	3	Chosen with the adviser from another course's list to broaden the specialisation.	-	TH category extension (6 cr); Mines free electives
Sem 8 10 cr total	**Graduation Design / Bachelor Thesis**	8	Independent project under supervision: literature review, data collection or design, analysis, thesis and oral defence.	-	TH Comprehensive Thesis Training (15 cr); CUGB BS Thesis (6 cr, 12 wks); MIT 12.TIP + 12.THU
	Frontiers Seminar of the Discipline	2	Guest lectures from industry and research institutes on new technologies, digitalisation and sustainability.	-	CUGB Frontiers of the Discipline (S7); MIT 12.00

6. Mineral Engineering course

Responsible unit: Institute of Mineral Engineering (mineralogy and petrology, mineral material development, rock application divisions). Graduate profile: Mineral engineers who characterise minerals and rocks, process them, and develop industrial mineral, stone, aggregate and functional-material products.

Course-specific credits: Sem 4 12, Summer 2 4, Sem 5 20, Sem 6 20, Summer 3 5, Sem 7 18, Sem 8 10 = 89. Semesters 1-3 and the common part of Sem 4 (8 cr): see section B. Total degree: 164 credits.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Sem 4 12 cr total	**Optical Mineralogy and Petrography**	4	Polarising microscope, optical properties of minerals, identification of rock-forming minerals, rock textures.	Nesse, Introduction to Optical Mineralogy (Oxford)	CUGB Crystal Optics (S3) + Petrology (S4); Mines GEGN307
	Systematic Mineralogy	3	Silicate and non-silicate mineral groups, structures, chemistry, occurrence and uses.	Nesse, Introduction to Mineralogy (Oxford); Deer, Howie and Zussman, An Introduction to the Rock-Forming Minerals (Mineralogical Society)	CUGB Crystallography and Mineralogy (S3); Mines GEOL321
	Physical Chemistry	3	Thermodynamics, phase equilibria, solutions, kinetics, surface chemistry and colloids.	Atkins and de Paula, Atkins' Physical Chemistry (Oxford)	CUGB Physical Chemistry C (S3)
	Structural Geology	2	Stress and strain, faults, folds, foliations, shear zones, stereographic projection, structural analysis of maps and cross-sections; structural control of ore and traps.	Fossen, Structural Geology (Cambridge); Lisle, Geological Structures and Maps (Elsevier)	CUGB Structural Geology (S4); Mines GEOL309; MIT 12.113
Summer 2 4 cr total	**Geological Mapping Field School (4 weeks)**	4	Systematic mapping of a sedimentary-igneous-metamorphic terrain; measured sections; structural data; geological map, cross-sections and field report.	Lisle, Brabham and Barnes, Basic Geological Mapping (Wiley-Blackwell)	CUGB Zhoukoudian Mapping Field School (6 wks); Mines GEGN316 Field Geology (5 wks); MIT 12.115
Sem 5 20 cr total	**Mineral Physics and Crystal Chemistry**	3	Crystal structures, defects, solid solution, phase transformations, physical properties of minerals.	Putnis, An Introduction to Mineral Sciences (Cambridge)	Mines GEOL321 Mineralogy and Mineral Characterization
	Analytical Mineralogy	4	X-ray diffraction, SEM-EDS, electron microprobe, Raman and IR spectroscopy, thermal analysis, automated mineralogy.	Cullity and Stock, Elements of X-Ray Diffraction (Pearson); Reed, Electron Microprobe Analysis and SEM in Geology (Cambridge)	Mines GEOL321 (XRD, SEM labs)
	Mineral Processing I (Comminution and Physical Separation)	4	Sampling, crushing and grinding, screening and classification, gravity separation, mass balances.	Wills and Finch, Wills' Mineral Processing Technology (Butterworth-Heinemann)	CUGB Intro to Mining and Mineral Processing (S7); Mines Metallurgical and Materials Eng.
	Materials Science and Engineering	3	Atomic structure and bonding in materials, crystal defects, diffusion, phase diagrams, mechanical, thermal and electrical properties.	Callister and Rethwisch, Materials Science and Engineering: An Introduction (Wiley)	CUG-W / CUGB Materials Science and Engineering programmes
	Industrial Minerals and Rocks	3	Geology, specifications and markets of industrial minerals (clays, feldspar, silica, phosphate, graphite, limestone, gypsum).	Kogel et al. (eds), Industrial Minerals and Rocks (SME); Harben and Kuzvart, Industrial Minerals: A Global Geology (IMIL)	No direct equivalent in the five benchmark plans reviewed as a separate subject; CUG-W Materials Science and Chemistry
	Process Mineralogy	3	Mineral liberation, texture and grade-recovery, automated mineralogy for beneficiation and geometallurgy.	Becker, Wightman and Evans (eds), Process Mineralogy (JKMRC)	Mines GEGN404 Ore Microscopy (related)
Sem 6 20 cr total	**Mineral Processing II (Flotation and Separation)**	4	Surface chemistry, flotation reagents and circuits, magnetic and electrostatic separation, dewatering, tailings.	Wills and Finch (Butterworth-Heinemann); Fuerstenau and Han (eds), Principles of Mineral Processing (SME)	Mines Kroll Institute for Extractive Metallurgy (research link)
	Non-metallic Mineral Materials (Ceramics, Glass, Cement)	4	Raw materials, processing and properties of ceramics, refractories, glass and Portland cement; quality control.	Carter and Norton, Ceramic Materials: Science and Engineering (Springer); Taylor, Cement Chemistry (Thomas Telford)	CUGB School of Materials Science and Technology
	Rock Mechanics and Dimension Stone	3	Rock strength and durability, quarrying and processing of dimension stone, decay and conservation of stone.	Siegesmund and Snethlage (eds), Stone in Architecture (Springer); Hudson and Harrison (Elsevier)	TH 30040142 Rock Mechanics; Mines MNGN321
	Construction Aggregates and Building Stone Technology	3	Aggregate resources, crushing and grading, specifications, alkali-aggregate reaction, rock products testing.	Smith and Collis (eds), Aggregates: Sand, Gravel and Crushed Rock Aggregates for Construction Purposes (Geological Society)	TH 40030352 Building Materials
	Gemmology	3	Physical and optical properties of gems, identification instruments, synthetics and treatments, gem deposits.	O'Donoghue (ed.), Gems (Butterworth-Heinemann); Read, Gemmology (Butterworth-Heinemann)	CUG-W Gemological Institute; CUGB School of Gemology

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
	Introduction to Extractive Metallurgy	3	Pyrometallurgy, hydrometallurgy (leaching, solvent extraction) and electrometallurgy; process flowsheets.	Havlik, Hydrometallurgy: Principles and Applications (Woodhead); Rosenqvist, Principles of Extractive Metallurgy (Tapir)	Mines Kroll Institute; Mines Metallurgical and Materials Eng.
Summer 3 5 cr total	**Production Internship (5 weeks)**	5	Placement with an industry partner; daily log, supervisor assessment and internship report.	-	CUGB Production Field Trip (6 wks) + Mine-Base Training; TH 40030584 Construction Internship; TH 40041003 Production Internship
Sem 7 18 cr total	**Mineral Materials Development Project (capstone)**	3	Team project to develop a product from a local mineral or rock: characterisation, processing tests, properties, cost.	Gupta and Yan, Mineral Processing Design and Operations (Elsevier)	No direct equivalent in the five benchmark plans reviewed
	Functional Mineral Materials	3	Clays and nanoclays, zeolites, graphite and graphene, diatomite, mineral adsorbents, catalysts and fillers.	Bergaya and Lagaly (eds), Handbook of Clay Science (Elsevier)	No direct equivalent in the five benchmark plans reviewed
	Testing and Standards for Mineral Products	2	Sampling plans, physical and chemical testing, product specifications and certification (ISO, ASTM, EN, national standards).	Relevant ISO / ASTM / national standards	No direct equivalent in the five benchmark plans reviewed
	Environmental Mineralogy and Tailings Utilisation	3	Mineral-water interactions, secondary minerals, re-use of tailings and waste rock in construction materials.	Vaughan and Wogelius (eds), Environmental Mineralogy (EMU Notes in Mineralogy); Lottermoser, Mine Wastes (Springer)	MIT 12.104 (related)
	Plant Design and Process Economics	3	Flowsheet development, equipment sizing, plant layout, capital and operating cost estimation.	Mular, Halbe and Barratt (eds), Mineral Processing Plant Design, Practice, and Control (SME)	Mines EBGN321 Engineering Economics
	Professional English and Technical Communication	2	Reading literature in the discipline, technical report and paper writing, presentations, international standards vocabulary.	Day and Gastel, How to Write and Publish a Scientific Paper (Cambridge)	CUGB Professional English for Resource Prospecting (S7); Mines writing-intensive (WI) capstones
	Elective (from any course of the faculty)	2	Chosen with the adviser from another course's list to broaden the specialisation.	-	TH category extension (6 cr); Mines free electives
Sem 8 10 cr total	**Graduation Design / Bachelor Thesis**	8	Independent project under supervision: literature review, data collection or design, analysis, thesis and oral defence.	-	TH Comprehensive Thesis Training (15 cr); CUGB BS Thesis (6 cr, 12 wks); MIT 12.TIP + 12.THU
	Frontiers Seminar of the Discipline	2	Guest lectures from industry and research institutes on new technologies, digitalisation and sustainability.	-	CUGB Frontiers of the Discipline (S7); MIT 12.00

7. Geological Information course

Responsible unit: Laboratory of Space (geospatial information division) with the Institute of Resource Exploration. Graduate profile: Geo-information engineers who build geological databases, GIS and 3D models, process remote-sensing data and apply machine learning to exploration and hazards.

Course-specific credits: Sem 4 12, Summer 2 4, Sem 5 20, Sem 6 20, Summer 3 5, Sem 7 18, Sem 8 10 = 89. Semesters 1-3 and the common part of Sem 4 (8 cr): see section B. Total degree: 164 credits.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Sem 4 12 cr total	**Database Systems**	3	Relational model, SQL, normalisation, transactions, indexing; design of a drill-hole and sample database.	Silberschatz, Korth and Sudarshan, Database System Concepts (McGraw-Hill)	TH 20740063 Database Technology and Applications
	Data Structures and Algorithms	3	Lists, trees, graphs, hashing, sorting and searching, algorithm complexity, spatial indexing (quad-tree, R-tree).	Goodrich, Tamassia and Goldwasser, Data Structures and Algorithms in Python (Wiley)	TH 20740112 Data Structures and Algorithms
	Structural Geology	3	Stress and strain, faults, folds, foliations, shear zones, stereographic projection, structural analysis of maps and cross-sections; structural control of ore and traps.	Fossen, Structural Geology (Cambridge); Lisle, Geological Structures and Maps (Elsevier)	CUGB Structural Geology (S4); Mines GEOL309; MIT 12.113
	Surveying and Geodesy	3	Reference ellipsoids and datums, map projections, levelling, total station and GNSS surveys, coordinate transformations.	Ghilani and Wolf, Elementary Surveying (Pearson); Torge and Muller, Geodesy (De Gruyter)	TH 30030611 Surveying; CUGB Surveying A (S4); CUG-W Surveying and Mapping Eng.
Summer 2 4 cr total	**Geological Mapping Field School (4 weeks)**	4	Systematic mapping of a sedimentary-igneous-metamorphic terrain; measured sections; structural data; geological map, cross-sections and field report.	Lisle, Brabham and Barnes, Basic Geological Mapping (Wiley-Blackwell)	CUGB Zhoukoudian Mapping Field School (6 wks); Mines GEGN316 Field Geology (5 wks); MIT 12.115
Sem 5 20 cr total	**Geographic Information Systems**	4	Vector and raster models, spatial reference, topology, spatial analysis, network analysis, map algebra, GIS project workflow.	Longley, Goodchild, Maguire and Rhind, Geographic Information Science and Systems (Wiley)	Mines GEGN475 Applications of GIS; TH 20030212; CUG-W National Engineering Research Center of GIS
	Remote Sensing and Digital Image Processing	4	Radiometry, sensors, geometric and radiometric correction, enhancement, classification, change detection.	Lillesand, Kiefer and Chipman, Remote Sensing and Image Interpretation (Wiley); Jensen, Introductory Digital Image Processing (Pearson)	MIT 12.421; TH 40040913 Remote Sensing Principles and Methods; Mines GEOL470; CUG-W Remote Sensing Science and Technology
	Cartography and Geological Map Compilation	3	Map design, symbolisation, generalisation, geological map standards, legend and cross-section compilation, digital publishing.	Slocum et al., Thematic Cartography and Geovisualization (CRC); Lisle, Geological Structures and Maps (Elsevier)	No direct equivalent in the five benchmark plans reviewed as a separate subject; CUG-W Geography and Information Eng.
	Geostatistics	3	Regionalised variables, variogram modelling, ordinary and indicator kriging, conditional simulation, uncertainty.	Isaaks and Srivastava, An Introduction to Applied Geostatistics (Oxford); Chiles and Delfiner, Geostatistics (Wiley)	MIT 12.012 (statistics part)
	Software Engineering	3	Requirements, design patterns, version control, testing, agile development, documentation.	Sommerville, Software Engineering (Pearson)	CUG-W Software Engineering programme (Mech. Eng. and Electronic Information)
	Mineral Deposits and Exploration for Information Engineers	3	Main deposit types and exploration workflow: what data are collected, how they are used for targeting and resource models.	Robb, Introduction to Ore-Forming Processes (Wiley-Blackwell); Moon, Whateley and Evans (Blackwell)	Mines GEGN401 Mineral Deposits
Sem 6 20 cr total	**Spatial Databases and Geological Data Models**	3	Spatial data types and SQL (PostGIS), geological data standards (GeoSciML), metadata, data governance.	Shekhar and Chawla, Spatial Databases: A Tour (Prentice Hall); Obe and Hsu, PostGIS in Action (Manning)	Mines GEGN432 Geological Data Management
	3D Geological Modelling	4	Implicit and explicit modelling, wireframes, drill-hole databases, block models, visualisation with mining software.	Mallet, Geomodeling (Oxford); Pyrcz and Deutsch, Geostatistical Reservoir Modeling (Oxford)	No direct equivalent in the five benchmark plans reviewed
	Photogrammetry and UAV Mapping	3	Stereo geometry, aerial triangulation, structure-from-motion, UAV mission planning, DEM and orthophoto production.	Wolf, Dewitt and Wilkinson, Elements of Photogrammetry with Applications in GIS (McGraw-Hill)	CUG-W Surveying and Mapping / Remote Sensing programmes
	GNSS Positioning	2	Satellite orbits and signals, code and carrier-phase positioning, RTK and PPP, error sources, geodetic networks.	Hofmann-Wellenhof, Lichtenegger and Wasle, GNSS - Global Navigation Satellite Systems (Springer)	CUG-W Surveying and Mapping Eng.
	Spatial Analysis and Mineral Prospectivity Mapping	3	Evidence layers, weights of evidence, fuzzy logic, logistic regression, random forests for targeting; validation.	Bonham-Carter, GIS for Geoscientists (Pergamon); Carranza, Geochemical Anomaly and Mineral Prospectivity Mapping in GIS (Elsevier)	Mines GEGN475; CUG-W Resource and Environment Big Data Eng.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
	Machine Learning for Geoscience	3	Supervised and unsupervised learning, neural networks, CNNs for images, model validation, applications to logs, maps and imagery.	Geron, Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow (O'Reilly)	TH 30030592 AI and Engineering Applications, 40240532 Intro to Machine Learning; CUG-W Data Science and Big Data Technology
	WebGIS and Geospatial Application Development	2	Web mapping services (WMS/WFS), web map libraries, mobile field-data apps, APIs.	Fu, Web GIS: Principles and Applications (Esri Press)	No direct equivalent in the five benchmark plans reviewed
Summer 3 5 cr total	**Production Internship (5 weeks)**	5	Placement with an industry partner; daily log, supervisor assessment and internship report.	-	CUGB Production Field Trip (6 wks) + Mine-Base Training; TH 40030584 Construction Internship; TH 40041003 Production Internship
Sem 7 18 cr total	**Geological Information System Design (capstone)**	3	Team design and implementation of an information system for a mine, survey or geohazard area: database, 3D model, web portal.	Sommerville, Software Engineering (Pearson)	No direct equivalent in the five benchmark plans reviewed
	Big Data and Cloud Computing for Geoscience	3	Distributed storage and processing, cloud platforms, Earth-observation data cubes, parallel computing.	Kleppmann, Designing Data-Intensive Applications (O'Reilly)	CUG-W Resource and Environment Big Data Engineering programme; TH 10030022 Data Science
	Hyperspectral and Radar (SAR) Remote Sensing	3	Imaging spectroscopy and mineral mapping; SAR principles, InSAR for ground deformation.	Elachi and van Zyl, Introduction to the Physics and Techniques of Remote Sensing (Wiley); Woodhouse, Introduction to Microwave Remote Sensing (CRC)	MIT 12.421; Mines GEOL470
	Digital Mine and Smart Mining	2	Mine planning software, fleet management, sensors and IoT, digital twins, automation.	Darling (ed.), SME Mining Engineering Handbook (SME)	No direct equivalent in the five benchmark plans reviewed
	Geohazard Monitoring and Early-Warning Systems	3	Landslide, subsidence and debris-flow monitoring (GNSS, InSAR, sensors), thresholds and warning systems.	Highland and Bobrowsky, The Landslide Handbook (USGS Circular 1325)	TH 40040782 Engineering Disaster Science; MIT Natural Hazards research area
	Professional English and Technical Communication	2	Reading literature in the discipline, technical report and paper writing, presentations, international standards vocabulary.	Day and Gastel, How to Write and Publish a Scientific Paper (Cambridge)	CUGB Professional English for Resource Prospecting (S7); Mines writing-intensive (WI) capstones
	Elective (from any course of the faculty)	2	Chosen with the adviser from another course's list to broaden the specialisation.	-	TH category extension (6 cr); Mines free electives
Sem 8 10 cr total	**Graduation Design / Bachelor Thesis**	8	Independent project under supervision: literature review, data collection or design, analysis, thesis and oral defence.	-	TH Comprehensive Thesis Training (15 cr); CUGB BS Thesis (6 cr, 12 wks); MIT 12.TIP + 12.THU
	Frontiers Seminar of the Discipline	2	Guest lectures from industry and research institutes on new technologies, digitalisation and sustainability.	-	CUGB Frontiers of the Discipline (S7); MIT 12.00

8. Space course

Responsible unit: Laboratory of Space (space science, earth atmosphere, geospatial information divisions). Graduate profile: Space and atmospheric scientists and engineers who can work on Earth-observation satellites, orbital and spacecraft systems, weather and climate, and space weather.

Course-specific credits: Sem 4 12, Summer 2 4, Sem 5 20, Sem 6 20, Summer 3 5, Sem 7 18, Sem 8 10 = 89. Semesters 1-3 and the common part of Sem 4 (8 cr): see section B. Total degree: 164 credits.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Sem 4 12 cr total	**Introduction to Space Science and Astronomy**	3	Solar system, stars and galaxies, telescopes and observation, the Sun-Earth system, history of space exploration.	Chaisson and McMillan, Astronomy Today (Pearson)	MIT 12.410J Observational Techniques of Optical Astronomy; MIT Astronomy minor
	Electromagnetism and Electromagnetic Waves	3	Maxwell's equations, wave propagation, polarisation, antennas basics, propagation in plasma and atmosphere.	Griffiths, Introduction to Electrodynamics (Cambridge)	No direct equivalent in the five benchmark plans reviewed (physics-department subject)
	Signals, Systems and Digital Signal Processing	3	Continuous and discrete signals, convolution, Fourier and z-transforms, sampling, filters, spectral analysis of geophysical time series.	Oppenheim, Willsky and Nawab, Signals and Systems (Pearson); Gubbins, Time Series Analysis and Inverse Theory for Geophysicists (Cambridge)	MIT 12.012 Statistics, Regression, Signal Processing
	Atmospheric Physics	3	Composition and structure of the atmosphere, thermodynamics, cloud microphysics, radiation basics, boundary layer.	Wallace and Hobbs, Atmospheric Science: An Introductory Survey (Academic Press)	MIT 12.003 Intro to Atmosphere, Ocean and Climate Dynamics; Mines GEOC407 Atmosphere, Weather and Climate
Summer 2 4 cr total	**Observatory and Meteorological Station Practice (4 weeks)**	4	Optical observation, radiosonde and surface weather observation, GNSS station operation, data reduction.	Station and observatory manuals	MIT Wallace Astrophysical Observatory practice; MIT 12.307 Weather and Climate Laboratory
Sem 5 20 cr total	**Orbital Mechanics**	4	Two-body problem, Kepler orbits, orbital elements, manoeuvres and transfers, perturbations, Earth-observation orbits.	Curtis, Orbital Mechanics for Engineering Students (Butterworth-Heinemann); Vallado, Fundamentals of Astrodynamics and Applications (Microcosm)	No direct equivalent in the five benchmark plans reviewed (aeronautics departments)
	Atmospheric Dynamics	4	Equations of motion on a rotating Earth, geostrophic and thermal wind, vorticity, waves, general circulation.	Holton and Hakim, An Introduction to Dynamic Meteorology (Academic Press)	MIT 12.003; MIT Atmospheres, Oceans and Climate programme
	Physics of Remote Sensing	4	Radiative transfer, spectral signatures, optical, thermal and microwave sensors, calibration, atmospheric correction.	Elachi and van Zyl, Introduction to the Physics and Techniques of Remote Sensing (Wiley); Petty, A First Course in Atmospheric Radiation (Sundog)	MIT 12.421 Remote Sensing: Physical Principles and Techniques
	Space Physics (Sun, Magnetosphere, Ionosphere)	4	Solar wind, magnetosphere, radiation belts, ionosphere, geomagnetic storms, plasma basics.	Kivelson and Russell (eds), Introduction to Space Physics (Cambridge)	MIT Planetary Science programme (related)
	Atmospheric Chemistry	2	Ozone, aerosols, greenhouse gases, air pollution chemistry and measurement from satellites.	Jacob, Introduction to Atmospheric Chemistry (Princeton); Seinfeld and Pandis, Atmospheric Chemistry and Physics (Wiley)	MIT 12.335 Experimental Atmospheric Chemistry
	Planetary Science	2	Formation of the solar system, planetary interiors, surfaces and atmospheres, small bodies, exoplanets.	de Pater and Lissauer, Planetary Sciences (Cambridge)	MIT 12.002; Mines GEOL410 Planetary Geology
Sem 6 20 cr total	**Spacecraft Systems Engineering**	4	Mission and systems engineering, power, thermal, attitude determination and control, structures, payloads, on-board computers.	Fortescue, Swinerd and Stark (eds), Spacecraft Systems Engineering (Wiley); Wertz, Everett and Puschell (eds), Space Mission Engineering: The New SMAD (Microcosm)	No direct equivalent in the five benchmark plans reviewed (aeronautics departments)
	Satellite Communications and Telemetry	3	Link budgets, modulation and coding, ground stations, telemetry, tracking and command.	Maral, Bousquet and Sun, Satellite Communications Systems (Wiley)	No direct equivalent in the five benchmark plans reviewed
	Weather Analysis and Numerical Weather Prediction	3	Synoptic analysis, numerical models, parameterisation, ensemble forecasting, verification.	Warner, Numerical Weather and Climate Prediction (Cambridge)	MIT 12.307 Weather and Climate Laboratory
	Climate System Science	3	Energy balance, atmosphere-ocean-land interactions, carbon cycle, climate variability and change, models.	Hartmann, Global Physical Climatology (Elsevier)	MIT 12.004 Intro to the Global Climate Cycle, 12.100 Plate Tectonics and Climate; TH Dept. of Earth System Science
	Satellite Meteorology	3	Meteorological satellites, imagery interpretation, sounding retrievals, precipitation and cloud products.	Kidder and Vonder Haar, Satellite Meteorology: An Introduction (Academic Press)	MIT 12.421 (related)
	Satellite Geodesy and GNSS	2	Satellite orbits, GNSS, satellite gravimetry and altimetry, reference frames.	Seeber, Satellite Geodesy (De Gruyter); Hofmann-Wellenhof et al., GNSS (Springer)	CUG-W Surveying and Mapping Eng.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
	Remote Sensing Image Processing	2	Pre-processing, enhancement, classification, object-based analysis, accuracy assessment.	Richards, Remote Sensing Digital Image Analysis (Springer)	TH 40040913; CUG-W Remote Sensing Science and Technology
Summer 3 5 cr total	**Production Internship (5 weeks)**	5	Placement with an industry partner; daily log, supervisor assessment and internship report.	-	CUGB Production Field Trip (6 wks) + Mine-Base Training; TH 40030584 Construction Internship; TH 40041003 Production Internship
Sem 7 18 cr total	**Space Mission Design (CubeSat capstone)**	4	Team design of a small-satellite Earth-observation mission: requirements, orbit, payload, subsystems, budget, review.	NASA, CubeSat 101: Basic Concepts and Processes for First-Time CubeSat Developers; Wertz et al., The New SMAD (Microcosm)	TH Dept. of Earth System Science satellite (launched 2024) - research link
	Planetary Geology and Space Resources	3	Impact cratering, volcanism and tectonics on other bodies, lunar and asteroid resources, in-situ resource utilisation.	Melosh, Planetary Surface Processes (Cambridge); Badescu (ed.), Moon: Prospective Energy and Material Resources (Springer)	Mines GEOL410 Planetary Geology; Mines Space Resources graduate programme
	Space Weather	2	Solar flares and CMEs, geomagnetic storms, effects on satellites, GNSS and power grids, forecasting.	Moldwin, An Introduction to Space Weather (Cambridge)	No direct equivalent in the five benchmark plans reviewed
	Microwave and SAR Remote Sensing	3	Radiometers, scatterometers, SAR imaging, polarimetry, InSAR applications.	Woodhouse, Introduction to Microwave Remote Sensing (CRC); Ulaby and Long, Microwave Radar and Radiometric Remote Sensing (Univ. of Michigan Press)	MIT 12.421
	Data Assimilation	2	Optimal interpolation, 3D/4D-Var, Kalman filters, assimilation of satellite observations.	Kalnay, Atmospheric Modeling, Data Assimilation and Predictability (Cambridge)	MIT Atmospheres, Oceans and Climate graduate programme
	Professional English and Technical Communication	2	Reading literature in the discipline, technical report and paper writing, presentations, international standards vocabulary.	Day and Gastel, How to Write and Publish a Scientific Paper (Cambridge)	CUGB Professional English for Resource Prospecting (S7); Mines writing-intensive (WI) capstones
	Elective (from any course of the faculty)	2	Chosen with the adviser from another course's list to broaden the specialisation.	-	TH category extension (6 cr); Mines free electives
Sem 8 10 cr total	**Graduation Design / Bachelor Thesis**	8	Independent project under supervision: literature review, data collection or design, analysis, thesis and oral defence.	-	TH Comprehensive Thesis Training (15 cr); CUGB BS Thesis (6 cr, 12 wks); MIT 12.TIP + 12.THU
	Frontiers Seminar of the Discipline	2	Guest lectures from industry and research institutes on new technologies, digitalisation and sustainability.	-	CUGB Frontiers of the Discipline (S7); MIT 12.00

9. Tourist District Development course

Responsible unit: Institute of Civil Engineering (tourist district development division). Graduate profile: Planners and engineers who assess landscapes and geoheritage, prepare tourist-district master plans, design facilities and infrastructure and manage destinations sustainably.

Course-specific credits: Sem 4 12, Summer 2 4, Sem 5 20, Sem 6 20, Summer 3 5, Sem 7 18, Sem 8 10 = 89. Semesters 1-3 and the common part of Sem 4 (8 cr): see section B. Total degree: 164 credits.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Sem 4 12 cr total	**Geomorphology and Landscape Geology**	3	Landforms and processes: fluvial, karst, glacial, coastal, volcanic and aeolian landscapes; landscape evolution and scenic value.	Huggett, Fundamentals of Geomorphology (Routledge)	MIT 12.163 Geomorphology; Mines GEGN203 Engineering Terrain Analysis
	Principles of Tourism and Geotourism	3	Tourism systems, demand and supply, tourist behaviour, nature-based tourism and geotourism concepts.	Goeldner and Ritchie, Tourism: Principles, Practices, Philosophies (Wiley); Dowling and Newsome (eds), Geotourism (Elsevier)	No direct equivalent in the five benchmark plans reviewed
	Urban and Regional Planning Principles	3	Land-use planning, zoning, regional development, infrastructure planning, planning process and public participation.	Levy, Contemporary Urban Planning (Routledge)	TH 30020572 Principles of Urban Planning; TH 40030832 Urban Planning and Transportation
	Engineering Surveying	3	Levelling, angle and distance measurement, total station, traversing, GNSS, topographic surveys, setting-out of works.	Ghilani and Wolf, Elementary Surveying: An Introduction to Geomatics (Pearson)	TH 30030611 Surveying + 40030402 Surveying Practice; CUGB Surveying A (S4)
Summer 2 4 cr total	**Landscape and Geoheritage Field Survey (4 weeks)**	4	2 weeks surveying camp; 2 weeks inventory of geosites, landscapes and tourism resources in a candidate district.	Reynard and Brilha (eds), Geoheritage: Assessment, Protection, and Management (Elsevier)	TH 40030402 Surveying Practice
Sem 5 20 cr total	**Tourism Planning**	4	Destination analysis, demand forecasting, zoning of tourist districts, tourism master-plan methods, stakeholder planning.	Gunn and Var, Tourism Planning: Basics, Concepts, Cases (Routledge); Inskeep, Tourism Planning: An Integrated and Sustainable Development Approach (Wiley)	No direct equivalent in the five benchmark plans reviewed
	Geoheritage Assessment and Geopark Management	3	Geodiversity, geosite inventory and quantitative assessment, protection, UNESCO Global Geopark criteria and management.	Gray, Geodiversity: Valuing and Conserving Abiotic Nature (Wiley-Blackwell); Reynard and Brilha (eds), Geoheritage (Elsevier)	No direct equivalent in the five benchmark plans reviewed; CUG-W campus Shaw Museum is a national 4A tourist site (practical example)
	Landscape Architecture and Site Design	4	Site analysis, landscape design principles, grading and drainage, planting, trails and viewpoints.	Booth, Basic Elements of Landscape Architectural Design (Waveland); Motloch, Introduction to Landscape Design (Wiley)	TH 40030862 Civil and Ecological Engineering
	Architectural Design Basics	3	Form, space and order, design drawing, small-building design studio.	Ching, Architecture: Form, Space and Order (Wiley)	TH 40030073 Architectural Design (1); TH 40030721 Intro to Architectural Design
	Environmental Impact Assessment	3	EIA process, screening and scoping, baseline studies, impact prediction, mitigation and monitoring, public consultation.	Glasson and Therivel, Introduction to Environmental Impact Assessment (Routledge)	Mines Environmental Geosciences track
	GIS for Planning	3	Land suitability analysis, accessibility and network analysis, 3D visualisation of planning proposals.	Longley et al., Geographic Information Science and Systems (Wiley)	Mines GEGN475; TH 20030212
Sem 6 20 cr total	**Tourism Infrastructure Engineering**	4	Access roads, trails, bridges and boardwalks, water supply and sanitation, power and telecom for tourist districts.	Garber and Hoel, Traffic and Highway Engineering (Cengage); Mays, Water Resources Engineering (Wiley)	TH 40030492 Road Engineering, 40030782 Bridge Engineering
	Tourism Economics and Market Analysis	3	Tourism demand and supply, pricing, economic impact and multipliers, market research and segmentation.	Stabler, Papatheodorou and Sinclair, The Economics of Tourism (Routledge)	TH 40910052 Urban and Real Estate Economics
	Tourist Facility Design	3	Design of visitor centres, accommodation, viewpoints and amenities; building codes and accessibility.	Neufert, Architects' Data (Wiley-Blackwell); Ching, Building Construction Illustrated (Wiley)	TH 30030522 Building Architecture
	Geohazards and Visitor Safety	3	Rockfall, landslide, flood, karst and volcanic hazards at tourist sites; risk assessment and safety management.	Keller, Environmental Geology (Pearson); Highland and Bobrowsky, The Landslide Handbook (USGS)	TH 00030052 Disasters and Countermeasures; Mines GEGN468
	Natural and Cultural Heritage Conservation	3	Heritage values, World Heritage system, conservation planning, protected-area management.	UNESCO, Operational Guidelines for the Implementation of the World Heritage Convention	No direct equivalent in the five benchmark plans reviewed
	Destination Management and Marketing	2	Destination management organisations, branding, visitor management, digital marketing.	Morrison, Marketing and Managing Tourism Destinations (Routledge)	No direct equivalent in the five benchmark plans reviewed
	Structures for Leisure Facilities	2	Structural concepts for pavilions, footbridges, towers and canopies; timber, steel and concrete options.	Ching, Onouye and Zuberbuhler, Building Structures Illustrated (Wiley)	TH 00030112 Structural Aesthetics

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Summer 3 5 cr total	**Production Internship (5 weeks)**	5	Placement with an industry partner; daily log, supervisor assessment and internship report.	-	CUGB Production Field Trip (6 wks) + Mine-Base Training; TH 40030584 Construction Internship; TH 40041003 Production Internship
Sem 7 18 cr total	**Tourist District Master Plan Studio (capstone)**	5	Team master plan for a real district: resource inventory, zoning, infrastructure, facilities, phasing, financial and environmental appraisal.	Gunn and Var, Tourism Planning (Routledge); Inskeep, Tourism Planning (Wiley)	TH 40031184 Integrated Course Design (analogue)
	Sustainable Tourism and Carrying Capacity	3	Sustainability principles, carrying capacity and limits of acceptable change, visitor impact monitoring.	Weaver, Sustainable Tourism (Routledge)	No direct equivalent in the five benchmark plans reviewed
	Tourism Law, Policy and Investment	2	Tourism legislation, land tenure, public-private partnerships, investment incentives.	Hall, Tourism Planning: Policies, Processes and Relationships (Pearson)	No direct equivalent in the five benchmark plans reviewed
	Heritage Interpretation and Visitor-Centre Design	2	Principles of interpretation, signage and trails, exhibition design for geological and natural sites.	Ham, Interpretation: Making a Difference on Purpose (Fulcrum)	No direct equivalent in the five benchmark plans reviewed
	Project Finance and Feasibility Studies	2	Market and financial feasibility, cash-flow models, risk analysis, financing structures for tourism projects.	Newnan, Eschenbach and Lavelle, Engineering Economic Analysis (Oxford)	TH 30030352 Engineering Economics; Mines EBGN321
	Professional English and Technical Communication	2	Reading literature in the discipline, technical report and paper writing, presentations, international standards vocabulary.	Day and Gastel, How to Write and Publish a Scientific Paper (Cambridge)	CUGB Professional English for Resource Prospecting (S7); Mines writing-intensive (WI) capstones
	Elective (from any course of the faculty)	2	Chosen with the adviser from another course's list to broaden the specialisation.	-	TH category extension (6 cr); Mines free electives
Sem 8 10 cr total	**Graduation Design / Bachelor Thesis**	8	Independent project under supervision: literature review, data collection or design, analysis, thesis and oral defence.	-	TH Comprehensive Thesis Training (15 cr); CUGB BS Thesis (6 cr, 12 wks); MIT 12.TIP + 12.THU
	Frontiers Seminar of the Discipline	2	Guest lectures from industry and research institutes on new technologies, digitalisation and sustainability.	-	CUGB Frontiers of the Discipline (S7); MIT 12.00

10. Petroleum Exploration and Engineering course

Responsible unit: Institute of Petrol (prospecting, mining engineering, mining machine, processing divisions). Graduate profile: Petroleum geoscientists and engineers covering exploration, drilling, reservoir and production engineering, equipment and refining.

Course-specific credits: Sem 4 12, Summer 2 4, Sem 5 20, Sem 6 20, Summer 3 5, Sem 7 18, Sem 8 10 = 89. Semesters 1-3 and the common part of Sem 4 (8 cr): see section B. Total degree: 164 credits.

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Sem 4 12 cr total	**Sedimentology and Stratigraphy**	4	Clastic and carbonate depositional systems, facies analysis, sequence stratigraphy, reservoir architecture.	Nichols, Sedimentology and Stratigraphy (Wiley-Blackwell); Boggs, Principles of Sedimentology and Stratigraphy (Pearson)	Mines GEOL314 Stratigraphy, GEOL315 Sedimentology and Stratigraphy; MIT 12.110A/B
	Structural Geology (Petroleum Applications)	3	Faults, folds, fractures and salt tectonics; trap geometry, fault seal, structural cross-sections and seismic structure.	Fossen, Structural Geology (Cambridge)	Mines GEOL308 Intro Applied Structural Geology, GEOL309
	Engineering Thermodynamics and Heat Transfer	3	Laws of thermodynamics, cycles of engines and compressors, conduction, convection, downhole temperature.	Cengel and Boles, Thermodynamics: An Engineering Approach (McGraw-Hill)	Mines GEGN330 Geoscientists Thermodynamics
	Sedimentary Petrology	2	Sandstone and carbonate petrography, diagenesis, porosity evolution, reservoir quality.	Boggs, Petrology of Sedimentary Rocks (Cambridge)	Mines GEGN307 Petrology
Summer 2 4 cr total	**Geological Mapping Field School (4 weeks)**	4	Systematic mapping of a sedimentary-igneous-metamorphic terrain; measured sections; structural data; geological map, cross-sections and field report.	Lisle, Brabham and Barnes, Basic Geological Mapping (Wiley-Blackwell)	CUGB Zhoukoudian Mapping Field School (6 wks); Mines GEGN316 Field Geology (5 wks); MIT 12.115
Sem 5 20 cr total	**Petroleum Geology**	4	Source rocks, generation and migration, reservoirs, seals and traps, petroleum systems, major basins of the world.	Gluyas and Swarbrick, Petroleum Geoscience (Wiley-Blackwell); Selley and Sonnenberg, Elements of Petroleum Geology (Academic Press)	Mines GEGN438 Petroleum Geology; TH 40040892 Marine Resources and Energy; CUG-W Petroleum Engineering programme
	Organic Geochemistry	2	Kerogen types, maturation, biomarkers, source-rock evaluation, hydrocarbon geochemistry for petroleum exploration.	Tissot and Welte, Petroleum Formation and Occurrence (Springer); Killops and Killops, Introduction to Organic Geochemistry (Blackwell)	CUGB Geochemistry incl. organic geochemistry (S5)
	Seismic Exploration	3	Elastic waves, reflection and refraction, acquisition geometry, sources and receivers, velocity analysis basics.	Sheriff and Geldart, Exploration Seismology (Cambridge)	MIT 12.210 Introduction to Seismology; Mines GEGN439 (seismic interpretation)
	Well Logging and Formation Evaluation	4	Electrical, nuclear, acoustic and imaging logs; porosity, saturation and lithology interpretation; logs in mineral boreholes.	Ellis and Singer, Well Logging for Earth Scientists (Springer); Rider and Kennedy, The Geological Interpretation of Well Logs (Rider-French)	Mines GEGN438 Petroleum Geology (log lab)
	Reservoir Rock and Fluid Properties	4	Porosity, permeability, capillary pressure, relative permeability, wettability; PVT behaviour of oil, gas and water.	Tiab and Donaldson, Petrophysics (Gulf); McCain, The Properties of Petroleum Fluids (PennWell)	Mines Petroleum Engineering programme
	Sedimentary Basin Analysis	3	Basin formation mechanisms, subsidence and thermal history, burial history modelling, basin evaluation.	Allen and Allen, Basin Analysis: Principles and Application to Petroleum Play Assessment (Wiley-Blackwell)	Mines GEOL314 (geohistory analysis)
Sem 6 20 cr total	**Petroleum Drilling Engineering**	4	Rotary rig systems, drilling fluids, hydraulics, bits, drill-string design, casing and cementing, well control basics.	Bourgoyne, Millheim, Chenevert and Young, Applied Drilling Engineering (SPE); Mitchell and Miska, Fundamentals of Drilling Engineering (SPE)	Mines Petroleum Engineering programme; CUG-W Petroleum Engineering programme
	Reservoir Engineering	4	Volumetrics, material balance, drive mechanisms, decline curves, water influx, displacement theory.	Craft and Hawkins, Applied Petroleum Reservoir Engineering (Pearson); Dake, Fundamentals of Reservoir Engineering (Elsevier)	Mines Petroleum Engineering programme
	Production Engineering	4	Inflow performance, nodal analysis, artificial lift, well stimulation (acidising, hydraulic fracturing), surface facilities.	Economides, Hill, Ehlig-Economides and Zhu, Petroleum Production Systems (Pearson); Guo, Lyons and Ghalambor, Petroleum Production Engineering (Gulf)	Mines Petroleum Engineering programme
	Seismic Interpretation and Petroleum Exploration	3	Seismic stratigraphy, structural mapping, play and prospect evaluation, risk and volumetrics.	Bacon, Simm and Redshaw, 3-D Seismic Interpretation (Cambridge)	Mines GEGN439 Petroleum Exploration Design
	Petroleum Machinery and Equipment	3	Drilling rigs, pumps and compressors, wellhead and BOP equipment, artificial-lift equipment, pipelines.	Lyons and Plisga (eds), Standard Handbook of Petroleum and Natural Gas Engineering (Gulf)	CUGB Cooperation Base for Deep Drilling Equipment; TH 40040972 Offshore Petroleum Engineering
	Well Testing	2	Pressure transient analysis, drawdown and build-up tests, type curves, interference tests.	Horne, Modern Well Test Analysis (Petroway); Lee, Well Testing (SPE)	Mines Petroleum Engineering programme

Semester	Subject	Cr.	Syllabus (main topics)	Recommended textbooks	Most similar subjects at benchmark universities
Summer 3 5 cr total	**Production Internship (5 weeks)**	5	Placement with an industry partner; daily log, supervisor assessment and internship report.	-	CUGB Production Field Trip (6 wks) + Mine-Base Training; TH 40030584 Construction Internship; TH 40041003 Production Internship
Sem 7 18 cr total	**Petroleum Exploration and Development Design (capstone)**	3	Team project: integrate seismic, logs and core to define prospects, estimate resources, propose wells and a development concept with economics.	Gluyas and Swarbrick, Petroleum Geoscience (Wiley-Blackwell)	Mines GEGN439 Petroleum Exploration Design (= PEGN439)
	Enhanced Oil Recovery	3	Waterflooding design, gas injection, chemical and thermal EOR, CO2-EOR and storage.	Green and Willhite, Enhanced Oil Recovery (SPE); Lake, Enhanced Oil Recovery (SPE)	Mines Petroleum Engineering programme
	Petroleum Refining and Processing	3	Crude oil characterisation, distillation, cracking, reforming, hydrotreating, gas processing, petrochemicals.	Gary, Handwerk and Kaiser, Petroleum Refining: Technology and Economics (CRC); Speight, The Chemistry and Technology of Petroleum (CRC)	No direct equivalent in the five benchmark plans reviewed in the geoscience units reviewed (chemical engineering departments)
	Natural Gas and Unconventional Resources	3	Gas reservoir engineering, shale gas and oil, coalbed methane, tight reservoirs, hydraulic fracturing geomechanics.	Guo and Ghalambor, Natural Gas Engineering Handbook (Gulf); Zoback and Kohli, Unconventional Reservoir Geomechanics (Cambridge)	CUG-W Resource Exploration Eng. (new-energy class); CUGB School of Energy Resources
	Reservoir Simulation	2	Flow equations, finite-difference discretisation, black-oil models, history matching, use of commercial simulators.	Ertekin, Abou-Kassem and King, Basic Applied Reservoir Simulation (SPE)	Mines Petroleum Engineering programme
	Petroleum Economics and HSE	2	Fiscal systems and production-sharing contracts, project economics, risk; health, safety and environment management.	Johnston, International Petroleum Fiscal Systems and Production Sharing Contracts (PennWell); Mian, Project Economics and Decision Analysis (PennWell)	Mines EBGN321; TH 40031141 Construction HSE Management (HSE part)
	Professional English and Technical Communication	2	Reading literature in the discipline, technical report and paper writing, presentations, international standards vocabulary.	Day and Gastel, How to Write and Publish a Scientific Paper (Cambridge)	CUGB Professional English for Resource Prospecting (S7); Mines writing-intensive (WI) capstones
Sem 8 10 cr total	**Graduation Design / Bachelor Thesis**	8	Independent project under supervision: literature review, data collection or design, analysis, thesis and oral defence.	-	TH Comprehensive Thesis Training (15 cr); CUGB BS Thesis (6 cr, 12 wks); MIT 12.TIP + 12.THU
	Frontiers Seminar of the Discipline	2	Guest lectures from industry and research institutes on new technologies, digitalisation and sustainability.	-	CUGB Frontiers of the Discipline (S7); MIT 12.00

C. Notes for the next step

- Electives: each course leaves 2-4 elective credits in Sem 7. Offering every course's subjects as electives to the others lets students combine, for example, Drill Engineering with Water Wells (Hydrogeology) or Geological Information with Mineral Exploration.
- Shared teaching: subjects such as Petrology, Structural Geology, Ore Deposit Geology, Applied Geophysics, Rock Mechanics, GIS and Professional English appear in several courses and can be taught once per semester to joint classes.
- Field bases: the plan assumes one mapping field base (Summer 2), a drilling teaching site, a surveying camp and an observatory/meteorological station. CUG's Beidaihe-Zhoukoudian-Zigui system and Mines' 5-week field camp are the benchmarks.
- Textbooks: confirm the latest edition and availability in your library; where a textbook is costly, CUG and Tsinghua typically publish national textbooks in the same subject that can be used instead.
- Similar-subject references are drawn from the documents reviewed in the first report (Tsinghua 2023 training plan, CUG (Beijing) Resources Prospecting Engineering plan, CUG (Wuhan) programme lists, MIT EAPS 2026 degree track, Colorado School of Mines 2025-26 Geological Engineering catalogue). Subjects in Mines' or MIT's other departments (petroleum, geophysics, aeronautics) were not reviewed course by course and are referenced at programme level.

Sources of benchmark subjects

Tsinghua 2023 Civil, Hydraulic and Ocean Engineering training plan: tsinghua.edu.cn/jxjywj/bkzy2023/zxzy/10.pdf

CUG (Beijing) Resources Prospecting Engineering programme: sesr.cugb.edu.cn/c/2020/06/17/619374.shtml

CUG (Wuhan) programmes: eniec.cug.edu.cn/info/1012/1198.htm; Faculty of Engineering: engineering.cug.edu.cn/About/About.htm

MIT EAPS 2026 Earth Science degree track: eaps.mit.edu/wp-content/uploads/2026/04/2026-earth-science-degree-track.pdf

Colorado School of Mines BS Geological Engineering 2025-26: catalog.mines.edu/undergraduate/academics/degrees/geologicalengineering/